Formative 1 “PMS" with answers.

1.The term that refers to a position away from the midline is which of the following?

a. Posterior.
b. Lateral.
c. Anterior.
d. Medial.

2.Which of the following best describes the anatomical position?

a. Palms of the hands are facing forwards.
b. Legs are away from each other.
c. Body is lying on its back.
d. Arms are abducted.

3.Frontal planes are also called which of the following?

a. Axial.
b. Oblique.
c. Horizontal.
d. Coronal.

4.Abduction means which of the following?

a. Moving the limbs toward the midline.
b. Moving the limbs above the midline.
c. Moving the limbs below midline.
d. Moving the limbs away from midline.

5.The anatomical position is best described as a person….

a. Lying on their stomach, palms up.
b. Standing, facing forward, palms forward.
c. Lying on their back, palms facing down.
d. Standing facing sideways, palms facing thighs.

6.The opposite to superior is called which of the following?

a. Inferior.
b. Superior.

c. Proximal.
d. Posterior.

7.Dorsal means which of the following?
a. Anterior.
b. Inferior.
c. Posterior.
d. Superior.

8.Which of the followings sites of superficial fascia has adipose tissue?
a. Scrotum.
b. Gluteal region.
c. Penis.
d. Eyelids.

9.A vertical plane, dividing the body into unequal left and right parts is which of the following?
a. Parasagittal.
b. Horizontal.
c. Median.
d. Coronal.

10.What is the type of the muscles of the upper limb?
a. Skeletal involuntary.
b. Cardiac.
c. Smooth.
d. Skeletal voluntary.

11.Adduction of a limb is the movement in which of the followings?
a. Coronal plane.
b. Horizontal plane.
c. Parasagittal plane.
d. Median Sagittal plane.

12.Transverse planes can also be called which of the following?
a. Coronal.
b. Oblique.

c. Longitudinal.
d. Horizontal.

13.The muscle of the wall of the stomach is called which of the following?

a. Cardiac.
b. Smooth.
c. Skeletal voluntary.
d. Skeletal involuntary.

14.Which of the following is a term of relation?

a. Distal.
b. Sagittal.
c. Transverse.
d. Coronal.

15.Which of the following provides summary of the four types of body planes?

a. Two are vertical, one is horizontal, and one is any odd angle.
b. Two are horizontal, one is vertical, and one is any odd angle.
c. All planes are vertical.
d. All planes are horizontal.

16.External means which of the following?

a. Below.
b. Outside.
c. Inside.
d. Above.

17.Prone refers to a patient lying on which of the following?

a. Their anterior body surface.
b. Their back, face up.
c. On their right side.
d. On their left side.

18.The normal organization is which of the following?

a. System, cell, tissue, organ.

b. Tissue, cell, system, organ.
c. Cell, tissue, organ, system.
d. Organ, cell, tissue, system.

19. Anatomy means which of the following?
a. Study of relations of different parts of the body.
b. Study function of the body.
c. Study structure, function and relations of different parts of the body.
d. Study of structure of the body.

20. Which of the following terms would be used in the name of a muscle that moves the leg toward the body?
a. Flexor.
b. Abductor.
c. Extensor.
d. Adductor.

21. Which of the following is correct about what happens during flexion?
a. The bone moves away from the body.
b. The angle between bones is decreased.
c. The bone moves toward the center of the body.
d. The angle between bones is increased.

22. What is the main component of the deep fascia?
a. Muscular tissue.
b. Nervous tissue.
c. Adipose tissue.
d. Fibrous tissue.

23. What is the plane that divides the human body into anterior and posterior parts?
a. Transverse.
b. Sagittal.
c. Midsagittal.
d. Coronal.

24. A patient who is supine is lying on which of the following?
    a. On his back.
    b. On the left side.
    c. Face down.
    d. On the right side.

Answers to formative 1 “PMS"

1.➜(b) Lateral.
2.➜(a) Palms of the hands are facing forwards.
3.➜(d) Coronal.
4.➜(d) Moving the limbs away from midline.
5.➜(b) Standing, facing forward, palms forward.
6.➜(a) Inferior.
7.➜(c) Posterior.
8.➜(b) Gluteal region.
9.➜(a) Parasagittal.
10.➜(d) Skeletal voluntary.
11.➜(a) Coronal plane.
12.➜(d) Horizontal.
13.➜(b) Smooth.
14.➜(a) Distal.
15.➜(a) Two are vertical, one is horizontal, and one is any odd angle.
16.➜(b) Outside.
17.➜(a) Their anterior body surface.
18.➜(c) Cell, tissue, organ, system.
19.➜(c) Study structure, function and relations of different parts of the body.
20.➜(d) Adductor.
21.➜(b) The angle between bones is decreased.
22.➜(d) Fibrous tissue.
23.➜(d) Coronal.
24.➜(a) On his back.

Formative 2 (PMS) answered.

Formative 2 “PMS” with answers.

1.What is the function of Cholesterol molecules in the plasma membrane?

a. form proteins.
b. increase permeability.
c. destroy phospholipids.
d. restrict movement of phospholipids.

2.Which of the following is part of the small intestine?

a. Ascending colon.
b. Cecum.
c. Ilium.
d. Transverse colon.

3.Transmembrane proteins span ?

a. half the membrane.
b. one side of the membrane.
c. the entire bilayer.
d. none of these.

4.Plasma membrane thickness ranges from ?

a. 5-6 nm.
b. 7.5-10 nm.
c. 2-3 nm.
d. 12-15 nm.

5.Which of the following is one of stomach bed?

a. Pancreas.
b. Right kidney.
c. Gall bladder.
d. Liver.

6.Phospholipids in the membrane are arranged as which?

a. double layer.
b. randomly.

c. single layer.
d. triple layer.

7.Which of the following best describes the gall bladder?
a. Pear shaped sac.
b. lies on the liver.
c. Its capacity is 500 ml.
d. Secretes the bile.

8.Which of the following is part of the large intestine?
a. Jejunum.
b. Duodenum.
c. Vermiform appendix.
d. Ilium.

9.The plasma membrane model is called the ?
a. rigid wall model.
b. fluid mosaic model.
c. sandwich model.
d. unit membrane model.

10.Membrane carbohydrates are attached to ?
a. DNA and RNA.
b. cytoplasm.
c. lipids and proteins.
d. mitochondria.

11.Which of the following is a prokaryotic organism?
a. Bacteria.
b. Amoeba.
c. Paramecium.
d. Yeast.

12.What is the function of the nose?
a. Smell.
b. Taste.
c. Mastication.

d. Speech.

13. The plasma membrane model is called the?
    a. fluid mosaic model.
    b. unit membrane model.
    c. rigid wall model.
    d. sandwich model.

14. What is the length of the esophagus?
    a. 15 cm.
    b. 45 cm.
    c. 35 cm.
    d. 25 cm.

15. The foreign body enters which of the following?
    a. Left main bronchus.
    b. Right lung.
    c. Left lung.
    d. Right main bronchus.

16. Which of the following best describes the fundus of the stomach?
    a. Pear shaped.
    b. Usually full of gas.
    c. Bulges upward to the right of cardiac opening.
    d. Lies on transpyloric plane

17. Which of the following best describes the esophagus?
    a. Has cartilage in its wall.
    b. Begins at C 6.
    c. Ends at T 10.
    d. Passes through the diaphragm at T 12.

18. The third molar teeth erupt at what age of the following?
    a. 18.
    b. 12.
    c. 10.
    d. 35.

19.Which of the following is a gland related to the endocrine system?

a. Parotid.
b. Sebaceous.
c. Pituitary.
d. Liver.

20.Common bile and main pancreatic ducts open into which of the following?

a. Third part of the duodenum.
b. First part of the duodenum.
c. Fourth part of the duodenum.
d. Second part of the duodenum.

21.Which of the following is part of the stomach?

a. Ampulla.
b. Cervix.
c. Isthmus.
d. Fundus.

22.Integral proteins are ?

a. free in the cytoplasm.
b. loosely attached to the surface.
c. found only outside the cell.
d. embedded within the bilayer.

23.The beginning of the trachea is which level of the following?

a. C 6.
b. Base of the skull.
c. Sternal angle.
d. C 3.

24.The plasma membrane appears trilaminar by which?

a. phase contrast microscope.
b. light microscope.
c. electron microscope.
d. compound microscope.

25.Which of the following best describes the tongue?

a. Has taste papillae.
b. Has bones inside.
c. Covered by skin.
d. Fixed organ.

26.Which of the following glands is a salivary gland?

a. Parotid.
b. Liver.
c. Thyroid.
d. Pituitary.

27.The part of the pharynx behind the mouth is called which of the following?

a. Laryngopharynx.
b. Epiglottis.
c. Oropharynx.
d. Nasopharynx.

28.Which of the following is a gland related to the digestive system?

a. Pituitary.
b. Suprarenal.
c. Parotid.
d. Pancreas.

29.Which of the following is an organ of the respiratory part of respiratory system?

a. Alveoli.
b. Larynx.
c. Nose.
d. Trachea.

30.The end of the trachea is which level of the following?

a. Sternal angle.
b. C 3.
c. C 6.

d. Base of the skull.

Answers to formative 2 “PMS".

1.➜(d) restrict movement of phospholipids.
2.➜(c) Ilium.
3.➜(c) the entire bilayer.
4.➜(b) 7.5-10 nm.
5.➜(a) Pancreas.
6.➜(a) double layer.
7.➜(a) Pear shaped sac.
8.➜(c) Vermiform appendix.
9.➜(b) fluid mosaic model.
10.➜(c) lipids and proteins.
11.➜(a) Bacteria.
12.➜(a) Smell.
13.➜(a) fluid mosaic model.
14.➜(d) 25 cm.
15.➜(d) Right main bronchus.
16.➜(b) Usually full of gas.
17.➜(b) Begins at C 6.
18.➜(a) 18.
19.➜(c) Pituitary.
20.➜(d) Second part of the duodenum.
21.➜(d) Fundus.
22.➜(d) embedded within the bilayer.
23.➜(a) C 6.
24.➜(c) electron microscope.
25.➜(a) Has taste papillae.
26.➜(a) Parotid.
27.➜(c) Oropharynx.
28.➜(c) Parotid.
29.➜(a) Alveoli.
30.➜(a) Sternal angle.

Formative 3 “PMS" with answers.

1.Which of the following organelles is responsible for detoxification?

a. Lysosome.
b. SER.
c. Golgi complex.
d. Mitochondria.

2.Lysosomes are membranous organelles. What is the type of surrounding membrane?

a. Smooth unit membrane.
b. Double unit membrane.
c. Rough unit membrane.
d. Coated membrane.

3.Which of the following is the structure of the Golgi bodies?

a. Cisternae and vesicles.
b. Cisternae of smooth membrane only.
c. Rough endoplasmic reticulum.
d. Tubules of smooth membranes.

4.Which of the following is an example of anatomic end artery?

a. Central artery of the retina.
b. Gastric arteries.
c. Coronary arteries.
d. Genicular arteries.

5.Which one of the following is a part of the basal ganglia?

a. Caudate nucleus.
b. Thalamus.
c. Pons.
d. Spinal cord.

6.The cerebral hemispheres are joined together by which of the following?

a. Corpus callosum.

b. Vermis.
c. Velum.
d. Basilar groove.

7.Which of the following is a part of the peripheral nervous system?
a. Spinal nerves.
b. Spinal cord.
c. Brain.
d. Meninges.

8.Which one of the following is a part of the brainstem?
a. Cerebrum.
b. Spinal cord.
c. Basal ganglia.
d. Medulla oblongata.

9.Which one of the following is a part of the diencephalon?
a. Pons.
b. Thalamus.
c. Spinal cord.
d. Basal ganglia.

10.Which of the following is an example of tortuous arteries?
a. Coronary arteries.
b. Ascending aorta.
c. Aortic arch.
d. Pulmonary trunk.

11.Primary somatosensory area lies in which of the following?
a. Parietal lobe.
b. Temporal lobe.
c. Occipital lobe.
d. Frontal lobe.

12.The branches of the ascending aorta are which of the following?
a. Brachiocephalic artery.
b. Descending thoracic aorta.

c. Pulmonary arteries.
d. Coronary arteries.

13.What is the EM structure of the cell membrane?
a. Two glycocalyx coat layers.
b. Cholesterol bilayer.
c. Three layers of phospholipids.
d. Trilaminar structure.

14.The size of the Golgi depends on which of the following factors?
a. The activity of the cell in protein secretion.
b. The size of the cell.
c. The activity of the cell in amoeboid movement.
d. The size of the nucleus.

15.The cavity of the diencephalon is called which of the following?
a. Fourth ventricle.
b. Third ventricle.
c. Cerebral aqueduct.
d. Lateral ventricle.

16.A 25-year-old male complains of muscle weakness and enlarged liver. He was diagnosed as glycogen storage disease. Investigations reveal genetic deficiency of one of cell enzymes. Where is the site of the defect in this patient?
a. Ribosome.
b. Lysosome.
c. S.E.R.
d. Nucleus.

17.The cerebellum is connected to the pons by which of the following?
a. Middle cerebellar peduncle.
b. Cerebral peduncles.
c. Inferior cerebellar peduncle.
d. Superior cerebellar peduncle.

18.Which one of the following is a part of the brainstem?

a. Pons.
b. Basal ganglia.
c. Spinal cord.
d. Cerebrum.

19. What is the stain of smooth endoplasmic reticulum?
a. Acidophilic by H&E.
b. Silver stain.
c. PAS.
d. Basophilic by H&E.

20. What is the organelle responsible for lipid synthesis?
a. Outer mitochondrial membrane.
b. RER.
c. SER.
d. Golgi bodies.

21. The cerebellum is connected to the midbrain by which of the following?
a. Middle cerebellar peduncle.
b. Inferior cerebellar peduncle.
c. Cerebral peduncle.
d. Superior cerebellar peduncle.

22. Which of the following is a part of the central nervous system?
a. Cranial nerves.
b. Meninges.
c. Spinal cord.
d. Spinal nerves.

23. The central nervous system is surrounded by which of the following?
a. Pleura.
b. Peritoneum.
c. Meninges.
d. Pericardium.

24.Which organelle is involved in protein synthesis?

a. Golgi apparatus.
b. Rough endoplasmic reticulum.
c. Smooth endoplasmic reticulum.
d. Lysosome.

25.What is the name of the cavity of the midbrain?

a. Cerebral aqueduct.
b. Fourth ventricle.
c. Lateral ventricle.
d. Central canal.

26.A 14-year-old male presented with progressive fatigue and severe anemia. He was diagnosed as a genetic defect in protein synthesis. Which organelle is defective in this patient?

a. Smooth endoplasmic reticulum.
b. Cisternae of Golgi.
c. Single ribosomes.
d. Free polysomes.

27.Which of the following is a part of the brainstem?

a. Cerebellum.
b. Medulla oblongata.
c. Cerebrum.
d. Spinal cord.

28.Which one of the following lies within the skull?

a. Brain.
b. Spinal cord.
c. Spinal nerves.
d. Cranial nerves.

29.How many spinal nerves?

a. 8 pairs.
b. 5 pairs.
c. 12 pairs.
d. 31 pairs.

Answers to formative 3 “PMS".

1.➜(b) SER.
2.➜(a) Smooth unit membrane..
3.➜(a) Cisternae and vesicles.
4.➜(a) Central artery of the retina.
5.➜(a) Caudate nucleus.
6.➜(a) Corpus callosum.
7.➜(a) Spinal nerves.
8.➜(d) Medulla oblongata.
9.➜(b) Thalamus.
10.➜(a) Coronary arteries.
11.➜(a) Parietal lobe.
12.➜(d) Coronary arteries.
13.➜(d) Trilaminar structure.
14.➜(a) The activity of the cell in protein secretion.
15.➜(b) Third ventricle.
16.➜(b) Lysosome.
17.➜(a) Middle cerebellar peduncle.
18.➜(a) Pons
19.➜(a) Acidophilic by H&E.
20.➜(c) SER.
21.➜(d) Superior cerebellar peduncle.
22.➜(c) Spinal cord.
23.➜(c) Meninges.
24.➜(b) Rough endoplasmic reticulum.
25.➜(b) Fourth ventricle.
26.➜(d) Free polysomes..

27.➜(b) Medulla oblongata.
28.➜(b) Spinal cord.
29.➜(d) 31 pairs.

Formative 4 “PMS" with answers.

1. Apex of the heart arteries from which of the following?
    a. Left ventricle.
    b. Right ventricle.
    c. Left atrium.
    d. Right atrium.
2. What is the stain of smooth endoplasmic reticulum?
    a. Silver stain.
    b. Basophilic by H&E.
    c. PAS.
    d. Acidophilic by H&E.
3. Mitral valve separates between which of the following?
    a. Right ventricle and left ventricle.
    b. Right atrium and left ventricle.
    c. Left atrium and right atrium.
    d. Right atrium and right ventricle.
4. Sternal angle lies Opposite which of the following?
    a. 2nd costal cartilage.
    b. 1st costal cartilage.
    c. 4th costal cartilage.
    d. 3rd costal cartilage.

5. Superior vena cava opens into which of the following?
    a. Left ventricle.
    b. Right ventricle.
    c. Left atrium.
    d. Right atrium.
6. Which of the following best describes the right ventricle?
    a. Infundibulum is its rough.
    b. Its muscular ridges are called musculi pectinati.
    c. Forms the base of the heart.
    d. Pumps blood into the pulmonary trunk.
7. Which of the following is the structure of the Golgi bodies?
    a. Cisternae and vesicles.
    b. Cisternae of smooth membrane only.
    c. Tubules of smooth membranes.
    d. Rough endoplasmic reticulum.
8. Right border of the heart is formed by which of the following?
    a. Right ventricle.
    b. Right atrium.
    c. Left atrium.
    d. Left ventricle.
9. What is the origin of the lymph?
    a. Liver.

b. Venous system.

c. Heart.

d. Tissue fluid.

10. Lymph vessels are present in which of the following?

a. Internal ear.

b. Central nervous system.

c. Eyeball.

d. Dermis of the skin.

11. Tricuspid valve separates between which of the following?

a. Left atrium and right atrium.

b. Right atrium and right ventricle.

c. Right ventricle and left ventricle.

d. Left ventricle and left atrium.

12. A 14-year-old male presented with progressive fatigue and severe anemia. He was diagnosed as a genetic defect in protein synthesis. Which organelle is defective in this patient?

a. Smooth endoplasmic reticulum.

b. Single ribosomes.

c. Free polysomes.

d. Cisternae of Golgi.

13. Fossa ovalis lies in which of the following chambers of the heart?

a. Left ventricle.

b. Left atrium.

c. Right ventricle.

d. Right atrium.

14. A 25-year-old male complains of muscle weakness and enlarged liver. He was diagnosed as glycogen storage disease. Investigations reveal genetic deficiency of one of cell enzymes. Where is the site of the defect in this patient?

a. Ribosomes.

b. SER.

c. Lysosomes.

d. Nucleus.

15. What is the organelle responsible for lipid synthesis?

a. RER.

b. Golgi bodies.

c. SER.

d. Outer mitochondrial membrane.

16. Which of the following produces lymphocytes?

a. Liver.

b. Thymus gland.

c. Parotid gland.

d. Pancreas.

17. Lysosomes are membranous organelles. What is the type of surrounding membrane?

    a. Coated membrane.

    b. Rough unit membrane.

    c. Double unit membrane.

    d. Smooth unit membrane.

18. Which of the following organelles is responsible for detoxification?

    a. SER.

    b. Lysosomes.

    c. Mitochondria.

    d. Golgi complex.

19. Sternocostal surface of the heart is not formed by which of the following?

    a. Left atrium.

    b. Right atrium.

    c. Left ventricle.

    d. Right ventricle.

20. The size of the Golgi depends on which of the following factors?

    a. The size of the nucleus.

    b. The activity of the cell amoeboid movement.

    c. The size of the cell.

d. The activity of the cell in protein secretion.

21. Which organelle is involved in protein synthesis?

a. Golgi apparatus.

b. Smooth endoplasmic reticulum.

c. Lysosomes.

d. Rough endoplasmic reticulum.

22. Coronary sinus ends into which of the following?

a. Right ventricle.

b. Left ventricle.

c. Right atrium.

d. Left atrium.

23. Right coronary artery arises from which of the following?

a. Pulmonary trunk.

b. Ascending aorta.

c. Arch of aorta.

d. Descending aorta.

24. Which of the following is a branch of the ascending aorta?

a. Brachiocephalic artery.

b. Left subclavian artery.

c. Right subclavian artery.

d. Left coronary artery.

25. Pulmonary veins open into which of the following?

a. Right atrium.

b. Left atrium.

c. Right ventricle.

d. Left ventricle.

26. Thoracic trunk ends into which of the following?

a. Inferior vena cava.

b. Right venous angle.

c. Left venous angle.

d. Superior vena cava.

27. Base of the heart is formed mainly by which of the following?

a. Left ventricle.

b. Right ventricle.

c. Left atrium.

d. Right atrium.

28. What is the EM structure of the cell membrane?

a. Three layers of phospholipids.

b. Trilaminar structure.

c. Cholesterol bilayer.

d. Two glycocalyx coat layers.

29. Descending thoracic aorta is the continuation of which of the following?

a. Arch of aorta.

b. Ascending aorta.

c. Brachiocephalic artery.

d. Left subclavian artery.

30. Inferior vena cava opens into which of the following?

a. Left ventricle.

b. Right atrium .

c. Right ventricle.

d. Left atrium.

Answers to formative 4 (PMS).

1.➜(a) Left ventricle.

2.➜(d) Acidophilic by H&E.

3.➜(b) Left atrium and left ventricle.

4.➜(a) 2nd costal cartilage.

5.➜(d) Right atrium.

6.➜(d) Pumps blood into the pulmonary trunk.

7.➜(a) Cisternae and vesicles.

8.➜(b) Right atrium.

9.➜(d) Tissue fluid.

10.➜(d) Dermis of the skin.

11.➜(b) Right atrium and right ventricle.

12.➜(c) Free polysomes.

13.➜(d) Right atrium.

14.➜(c) Lysosomes.

15.➜(c) SER.

16.➜(b) Thymus gland.

17.➜(d) Smooth unit membrane.

18.➜(a) SER.

19.➜(a) Left atrium.

20.➜(d) The activity of the cell in protein secretion.

21.➜(d) Rough endoplasmic reticulum.

22.➜(c) Right atrium.

23.➜(d) Descending aorta.

24.➜(d) Left coronary artery.

25.➜(b) Left atrium.

26.➜(c) Left venous angle.

27.➜(c) Left atrium.

28.➜(b) Trilaminar structure.

29.➜(a) Arch of aorta.

30.➜(b) Right atrium.

Formative 5 "PMS" with answers.

Question 1

Not yet answered

Flag question

Following is secondary organ of the female genital system?

- a. Uterus
- b. Ovary
- c. Vas deferens
- d. Testis

Question 2

Not yet answered

Flag question

Microtubules are composed of which of these?

- a. DNA strands
- b. Keratin monomers
- c. Tubulin dimers
- d. Actin dimers

Question 3

Not yet answered

Flag question

What is the longest part of the male urethra?

- a. Penile
- b. Prostatic
- c. Membranous
- d. Isthmus

?

Question 4

Not yet answered

Flag question

Proteasomes degrade proteins that are labeled with?

- a. Keratin
- b. Ubiquitin
- c. Tubulin
- d. Actin

Question 5

Not yet answered

Flag question

Which of the following describes the ductus difference?

- a. 20 cm long
- b. Continuation of the duct of the epididymis
- c. Joins duct of seminal vesicle to form the prostatic urethra
- d. Artery of ductus deferens arises from the testicular artery

Question 6

Not yet answered

Flag question

Failure of proteasomes contributes to which disorder?

- a. Diabetes
- b. Asthma
- c. Osteoporosis
- d. Parkinson's disease

Question 7

Not yet answered

Flag question

The length of the female urethra is which of the following?

- a. 45 cm
- b. 25 cm
- c. 10 cm
- d. 4 cm

Question 8

Not yet answered

Flag question

What is the most dilatable part of the male urethra?

- a. Penile urethra
- b. Prostatic urethra
- c. Isthmus urethra
- d. Membranous urethra

Question 9

Not yet answered

Flag question

What Proteasomes can destroy?

- a. DNA
- b. Lipids
- c. Misfolded proteins
- d. Mitochondria

Question 10

Not yet answered

Flag question

Which one of the following describes the uterus?

- a. Divided into fundus, body and ampulla.
- b. Solid organ.
- c. its normal position is anteverted anteflexed.
- d. 10 cm in length.

Question 11

Not yet answered

Flag question

Which of the following describes the left renal vein?

- a. Lies deep to the ureter
- b. Receives the left gonadal vein
- c. Shorter than the right renal vein
- d. Receives the azygos vein

Question 12

Not yet answered

Flag question

What one of the following is a part of the external genital organs in the male?

- a. Epididymis
- b. Penis
- c. Seminal vesicle
- d. Testis

Question 13

Not yet answered

Flag question

where Ribosomal proteins are synthesized ?

- a. Mitochondria
- b. Cytoplasmic ribosomes
- c. Nucleus
- d. Nucleolus

Question 14

Not yet answered

Flag question

Following is the primary organ of the female genital system?

- a. Vulva
- b. Vagina
- c. Ovary
- d. Uterus

Question 15

Not yet answered

Flag question

What is the function of the testis?

- a. Production of sperms
- b. Production of ova
- c. Secretion of progesterone
- d. Secretion of estrogen

Question 16

Not yet answered

Flag question

Microtubules are essential for which?

- a. Fat metabolism
- b. Mitotic spindle formation
- c. Blood glucose control
- d. DNA repair only

Question 17

Not yet answered

Flag question

Colchicine is used to which of these?

- a. Treat asthma
- b. Synthesize proteins
- c. Stop mitosis
- d. Increase ATP

Question 18

Not yet answered

Flag question

The normal position of the uterus is which of the following?

- a. Retroverted
- b. Retroflexed
- c. Anteverted anteflexed
- d. Retroverted retroflexed

Question 19

Not yet answered

Flag question

What is the function of the seminal vesicle?

- a. Secretion of testosterone
- b. Secretion of estrogen
- c. Secretion of fructose
- d. Secretion of progesterone

Question 20

Not yet answered

Flag question

Which of the following is the primary organ of male genital system?

- a. Seminal vesicle
- b. Testes
- c. Prostate
- d. Epididymis

Question 21

Not yet answered

Flag question

Cytoskeleton provides which of these?

- a. ATP
- b. Structural support and motility
- c. Lipid storage
- d. Digestion

Question 22

Not yet answered

Flag question

Union of duct of vas deference and duct of seminal vesicle gives which?

- a. Penile urethra.
- b. Epididymis.
- c. Ejaculatory duct.
- d. Prostate.

Question 23

Not yet answered

Flag question

Which of the following is the secondary organ of male genital system?

- a. Prostate
- b. Testes
- c. Vulva
- d. Ovary

Question 24

Not yet answered

Flag question

Polysomes refer to which?

- a. Single ribosome
- b. Ribosomes not involved in translation
- c. Ribosomes inside nucleus
- d. Several ribosomes attached to mRNA

Question 25

Not yet answered

Flag question

What of the following is part of the urinary system?

- a. Uterus
- b. Vulva
- c. Kidney
- d. Testes

Question 26

Not yet answered

Flag question

Which of the following is a part of the epididymis?

- a. Fundus
- b. Ampulla
- c. Isthmus
- d. Head

Question 27

Not yet answered

Flag question

The length of the ureter is which of the following?

- a. 10 cm
- b. 45 cm
- c. 25 cm
- d. 4 cm

Question 28

Not yet answered

Flag question

where Actin filaments are located mainly?

- a. Only in nucleus
- b. In lysosomes
- c. In mitochondria
- d. Beneath the cell membrane

Answers to formative 5 (PMS)

1.➜(b) Ovary.
2.➜(c) Tubulin dimers.
3.➜(a) Penile.
4.➜(b) Ubiquitin.
5.➜(b) Continuation of the duct of the epididymis.
6.➜(d) Parkinson's disease.
7.➜(d) 4 cm.
8.➜(b) Prostatic urethra.
9.➜(c) Misfolded proteins.
10.➜(c) its normal position is anteverted anteflexed.
11.➜(b) Receives the left gonadal vein.
12.➜(b) Penis.
13.➜(b) Cytoplasmic ribosomes.
14.➜(c) Ovary.
15.➜(a) Production of sperms.

16.➜(b) Mitotic spindle formation.
17.➜(c) Stop mitosis.
18.➜(c) Anteverted anteflexed.
19.➜(c) Secretion of fructose.
20.➜(b) Testes.
21.➜(b) Structural support and motility.
22.➜(c) Ejaculatory duct.
23.➜(b) Testes.
24.➜(d) Several ribosomes attached to mRNA.
25.➜(c) Kidney.
26.➜(d) Head.
27.➜(c) 25 cm.
28.➜(d) Beneath the cell membrane.

Formative 6 “PMS" with answers.

1.Which of the following is one of internal genital organs in the male?

a. Scrotum.
b. Testis.
c. Uterus.
d. Ovary.

_____

2.The condition involving iron overload, often from blood transfusions, is called?

a. Haemosiderosis.
b. Lipofuscinosis.
c. Hemochromatosis.
d. Melanosis.

_____

3.Union of duct of vas deference and duct of seminal vesicles gives which?

a. Ejaculatory duct.
b. Penile urethra.
c. Prostate.
d. Epididymis.

_____

4.Which of the following describes the thyroid gland?

a. Consists of 2 lobes and isthmus.
b. parathyroid glands Lie on its anterior surface.

c. Lies in the abdomen.
d. shorter than the right renal vein.

_____

5.Which of the following describes the uterine tube.

a. connects the vagina with the peritoneal cavity.
b. Shorter than the uterus.
c. Site of fertilization.
d. Fibrous tube.

_____

6.Glycogen is stored in large amounts in which cells?

a. Skin and lung cells.
b. Nerve and Kidney cells.
c. Liver and muscle cells.
d. All animal cells equally.

_____

7.Which of the following is one to the endocrine system organs?

a. Pancreas.
b. Suprarenal gland.
c. Liver.
d. Parotid gland.

_____

8.Which of the following is a part of the epididymis?

a. Body.
b. Ampulla.
c. Fundus.

d. Isthmus.

_____

9.What is the length of the ovary?

a. 25 cm.
b. 20 cm.
c. 5 cm.
d. 10 cm.

_____

10.Carotenoid pigments are an example of which of these?

a. Crystalline pigments.
b. Endogenous pigments.
c. Exogenous pigment.
d. Hematogenous pigments.

_____

11.The normal position of the uterus is which of the following?

a. Retroverted retroflexed.
b. Retroverted.
c. Retroflexed.
d. Anteverted anteflexed.

_____

12.Hemochromatosis is primarily caused by?

a. A lack of dietary iron.
b. Iron overload from transfusions.

c. Excessive breakdown of red blood cells.
d. Impaired iron absorption in the small intestine.

_____

13.Which of the following is a structure of the vulva?

a. Seminal vesicles.
b. Uterine tube.
c. Mons pubis.
d. Uterus.

_____

14.Which of the following located in the scrotum?

a. Testis.
b. Prostate.
c. Seminal vesicle.
d. Paraurethral glands.

_____

15.A fully engorged fat cell has a nucleus that is which of these?

a. Centrally located and spherical.
b. Eccentric and vesicular.
c. Multi-lobed and central.
d. Highly flattened and peripheral.

_____

16.The iron-containing pigment that results from hemoglobin breakdown is which of these?

a. Bilirubin.
b. Lipofuscin.

c. Melanin.
d. Haemosiderin.

_____

17.Which of the following is one to the endocrine system?

a. Liver.
b. Pituitary gland.
c. Pancreas.
d. Parotid gland.

_____

18.What is the shortest part of the male urethra?

a. Membranous.
b. Penile.
c. Prostatic.
d. Isthmus.

_____

19.Bile pigments result from the breakdown of which?

a. Red blood cells.
b. Muscle cells.
c. Melanocytes.
d. Fat cells.

_____

20.Lipofuscin is sometimes called "wear and tear" pigment because it?

a. Is involved in oxygen transport.

b. Gives skin its color.
c. Increases in amount with age.
d. Is derived from external sources.

_____

21.For light microscopic localization, glycogen can be stained with which stain?

a. silver stain.
b. Sudan black.
c. PAS or Best's carmine.
d. Perls Prussian blue.

_____

22.Which of the following is a part of the uterus?

a. Head.
b. Cervix.
c. Bulb.
d. Ampulla.

_____

23.A pigment present in muscle cells is which of these?

a. Myoglobin.
b. Hemoglobin.
c. Lipofuscin.
d. Melanin.

_____

24.What is the function of the seminal vesicle?

a. Secretion of estrogen.

b. Secretion of fructose.
c. Secretion of progesterone.
d. Secretion of testosterone.

Answers to formative 6 “PMS"

1.➜(b) Testis.

2.➜(a) Haemosiderosis.

3.➜(a) Ejaculatory duct.

4.➜(a) Consists of 2 lobes and isthmus.

5.➜(c) Site of fertilization.

6.➜(c) Liver and muscle cells.

7.➜(b) Suprarenal gland.

8.➜(a) Body.

9.➜(c) 5 cm.

10.➜(c) Exogenous pigments.

11.➜(d) Anteverted anteflexed.

12.➜(d) Impaired iron absorption in the small intestine.

13.➜(c) Mons pubis.

14.➜(a) Testis.

15.➜(d) Highly flattened and peripheral.

16.➜(d) Haemosiderin.

17.➜(b) Pituitary gland.

18.➜(a) Membranous.

19.➜(a) Red blood cells.

20.➜(c) Increases in amount with age.

21.➜(c) PAS or Best's carmine.

22.➜(b) Cervix.

23.➜(a) Myoglobin.

24.➜(b) Secretion of fructose.

Formative 7 “PMS”

1. The thickness of the P.M. is?

a. 7–10 μm
b. 1–3 nm
c. 70–100 μm
d. 7–10 nm

2. What is the second envelope of the body?

a. Bone
b. Muscles
c. Fascia
d. Skin

3. Which of the following is the longest part of the male urethra?

a. Penile
b. Spongy
c. Prostatic
d. Membranous

4. What is the polysome?

a. Membranous organelle
b. Organelle involved in glycogenolysis
c. A group of ribosomes attached by m-RNA
d. A group of ribosomes attached by t-RNA

5. What is the right statement concerning the movement of extension of the hip joint?

a. Move the lower limb away from the midline in the coronal plane
b. Move the lower limb posteriorly in the paramedian plane
c. Rotate the lower limb so that the anterior surface faces medially
d. Move the lower limb anteriorly in the paramedian plane

6. Where are ribosomes formed?

a. Golgi bodies
b. Nucleolus
c. R.E.R.
d. S.E.R.

7. What statement does not describe the skeletal muscles?

a. Motor point is at about the midpoint on the deep surface of the muscle
b. Attachment which moves the most is the insertion
c. Its nerve supply is mainly sensory
d. Degree of mobility between origin and insertion may be reversed

8. What is the right statement concerning the anatomic position?

a. The person has the ankles several inches apart
b. The person has the upper limbs by the sides of the trunk
c. The person is facing laterally
d. The person has the palms of the hands directed medially

9. Smooth endoplasmic reticulum?

a. Forms proteins
b. Is basophilic
c. Contains ribosomes
d. Is acidophilic

10. The process of pinocytosis means?

a. Cells engulfing solids
b. Cells engulfing fluids
c. Mitotic cell division
d. Cells excreting fluids

11. A 50-year-old female comes to a neurologist complaining of memory loss. Which of the following pigments increase in neurons of aged patients?

a. Tattoo pigment
b. Melanin pigment
c. Lipochromes pigment
d. Lipofuscin pigments

12. What is the function of free polysomes?

a. Synthesis of secretory proteins
b. Conduction of material in the cytoplasm
c. Formation of cytoplasmic proteins
d. Formation of steroids

13. Which organelle is involved in protein synthesis?

a. Golgi apparatus
b. Smooth endoplasmic reticulum
c. Rough endoplasmic reticulum

d. Lysosome

14. Which one of the following is an endocrine gland?

a. Parotid gland
b. Submandibular gland
c. Pituitary gland
d. Liver

15. Which statement best describes the left lung?

a. Has 3 lobes
b. Has cardiac notch
c. Related to superior vena cava
d. Has 2 fissures

16. Cell membrane has?

a. Two glycocalyx coat layers
b. Trilaminar structure
c. Cholesterol bilayer
d. Three layers of phospholipids

17. Which of these muscle types are found in the heart?

a. Skeletal
b. Smooth
c. Cardiac
d. Voluntary

18. Which of the following is the normal position of the uterus?

a. Anteverted anteflexed
b. Retroverted
c. Retroverted retroflexed
d. Retroflexed

19. What is the primary sexual organ of the female genital system?

a. Ovary
b. Vagina
c. Uterus
d. Vulva

20. Which of the following best describes the gall bladder?

a. Pear-shaped sac
b. Secretes the bile

c. Lies over the liver
d. Its capacity is 500 ml

21. The size of the Golgi depends on which factor?

a. The size of the nucleus
b. The size of the cell
c. The activity of the cell in amoeboid movement
d. The activity of the cell in protein secretion

22. A 14-year-old male with anemia due to a genetic defect in protein synthesis. Which organelle is responsible?

a. Single ribosomes
b. Free polysomes
c. Cisternae of Golgi
d. Smooth endoplasmic reticulum

23. The Golgi is formed of:

a. Rough endoplasmic reticulum
b. Cisternae and vesicles
c. Tubules of smooth membranes
d. Cisternae of smooth membrane only

24. What is the stain of smooth endoplasmic reticulum?

a. PAS
b. Basophilic by H&E
c. Silver stain
d. Acidophilic by H&E

25. They stabilize cell structure and resist tension:

a. Microtubules
b. Intermediate filaments
c. Myosin filaments
d. Microfilaments

26. Vessels that carry blood from the heart to the periphery are:

a. Sinusoids
b. Arteries
c. Veins
d. Capillaries

27. Ribosomes are formed in?

a. R.E.R.
b. S.E.R.
c. Nucleolus
d. Golgi

28. Which is moved the least during muscle contraction?

a. Ligaments
b. Insertion
c. Joints
d. Origin

29. Free polysomes are mainly concerned with?

a. Formation of steroids
b. Conduction of material in the cytoplasm
c. Formation of cytoplasmic proteins
d. Synthesis of secretory proteins

30. Which of the following is part of the respiratory system?

a. Stomach
b. Esophagus
c. Kidney
d. Trachea

31. Detoxification in cells is performed by?

a. Lysosome
b. S.E.R.
c. Golgi complex
d. Mitochondria

32. Lysosomes are membranous organelles. What is the type of surrounding membrane?

a. Double unit membrane
b. Rough unit membrane
c. Smooth unit membrane
d. Coated membrane

33. Sagittal (median) plane of the human body means which of the following?

a. The horizontal line dividing the body into upper and lower halves
b. The vertical line dividing the body into anterior and posterior halves
c. The vertical line dividing the body into unequal right and left halves
d. The vertical line dividing the body into equal right and left halves

34. Which of the following male organs is responsible for fructose production?

a. Testis
b. Kidneys
c. Seminal vesicles
d. Prostate gland

35. What is the organelle responsible for lipid synthesis?

a. Golgi bodies
b. Outer mitochondrial membrane
c. S.E.R.
d. R.E.R.

36. Where in the tracheobronchial tree is a common site for a foreign object to lodge?

a. Left tertiary bronchus
b. Beginning of the trachea
c. Right primary bronchus
d. Left primary bronchus

37. What is the secondary cartilaginous joint of the following?

a. Ankle joint
b. Shoulder joint
c. Sutures of the skull
d. Symphysis pubis

38. Where are red blood cells produced?

a. Bone marrow
b. Heart
c. Lymph nodes
d. Liver

39. Which of the following joints is synovial with hinge variety?

a. Sutures of the skull
b. Inferior tibiofibular joint
c. Elbow joint
d. Gomphosis

40. Identify the part of the respiratory system in which gas exchange takes place?

a. Alveoli
b. Pharynx
c. Larynx
d. Trachea

41. The central nervous system is surrounded by which of the following?

a. Meninges
b. Peritoneum
c. Pleura
d. Pericardium

ANSWERS

1. d

2. c

3. b

4. c

5. b

6. b

7. c

8. b

9. d

10. b

11. d

12. c

13. c

14. c

15. b

16. b

17. c

18. a

19. a

20. a

21. d

22. b

23. b

24. d

25. b

26. b

27. c

28. d

29. c

30. d

31. b

32. c

33. d

34. c

35. c

36. c

37. d

38. a

39. c

40. a

41. a

Formative 8 “PMS” with answers.

Prepared by group: “A"

Question 1
Which of the following is the definition of fetus?

· a. Period from the third week to the eighth week
· b. Period from fertilization to end of the second week
· c. Period from labor to end of the first month
· d. Period from the third month to the birth

Question 2
Which of the following is the age of childhood?

· a. Period from 16-18 years postnatal.
· b. Period from the second year postnatal to the twelfth year.
· c. Period from 14-16 years postnatal.
· d. Period from 19-21 years postnatal.

Question 3
Ampulla of the uterine tube is the site of which of the following?

· a. Ovulation
· b. Implantation
· c. Insemination
· d. Fertilization

Question 4
Which of the following is the definition of embryo?

· a. Period from fertilization to end of the second week
· b. Period from the third week to the eighth week
· c. Period from the third month to the birth
· d. Period from labor to end of the first month

Question 5
Infundibulum of the uterine tube opens into which of the following?

· a. Uterine cavity
· b. Vestibule
· c. Vagina
· d. Peritoneal cavity

Question 6
Down syndrome means which of the following?

· a. Loss of an X chromosome in a female
· b. Presence of extra X chromosome in male (47 XXY)
· c. Presence of a third copy of chromosome 21
· d. Deletion of small segment of chromosome 22

Question 7
The pars fibrosa of the nucleolus contains which?

· a. Dense lipid droplets
· b. Chromatin filaments and filamentous RNA
· c. Ribosomal subunits
· d. Only carbohydrates

Question 8
Where ribosomal assembly within the nucleolus occur?

· a. Nuclear pore
· b. Pars fibrosa
· c. Pars granulosa
· d. Fibrillar center

Question 9
Which of the following does not describe the primordial germ cells?

· a. Originated in the wall of the yolk sac
· b. Give rise to sperms in the male
· c. Migrated to reach their terminal site
· d. Give rise to ovary in the female

Question 10
Ovary is the site of which of the following?

· a. Implantation
· b. Insemination
· c. Ovulation
· d. Fertilization

Question 11
The exchange through nuclear pores is which of these?

· a. Selective for specific substances
· b. Only for ions
· c. Only for water
· d. Non-selective

Question 12
The nuclear envelope is visible under a light microscope as a?

· a. Light eosinophilic line
· b. Series of granules
· c. Clear space
· d. Dark basophilic membrane

Question 13
The only mammalian cells that are non-nucleated are?

· a. Hepatocytes
· b. Osteoclasts
· c. Plasma Cells
· d. Red Blood Cells

Question 14
Which of the following is the definition of puberty?

· a. Physical maturation
· b. Eruption of milk teeth
· c. Sexual maturation
· d. Eruption of permanent teeth

Question 15
Which of the following is the age of puberty in the male?

· a. Period from 19-21 years postnatal.
· b. Period from 16-18 years postnatal.
· c. Period from 14-16 years postnatal.
· d. Period from the second year postnatal to the twelfth year.

Question 16

Turner syndrome means which of the following?

· a. Deletion of small segment of chromosome 22
· b. Presence of a third copy of chromosome 21
· c. Loss of an X chromosome in a female
· d. Presence of extra X chromosome in male (47 XXY)

Question 17
Which of the following is the definition of newborn?

· a. Period from labor to end of the first month
· b. Period from fertilization to end of the second week
· c. Period from the third week to the eighth week
· d. Period from the third month to the birth

Question 18
Posterior fornix of vagina is the site of which of the following?

· a. Implantation
· b. Fertilization
· c. Ovulation
· d. Insemination

Question 19
The nuclear envelope is formed of which?

· a. A single lipid layer
· b. A non-selective barrier
· c. Two parallel unit membranes
· d. Continuous membrane without pores

Question 20
Klinefelter syndrome means which of the following?

· a. Deletion of small segment of chromosome 22
· b. Presence of a third copy of chromosome 21
· c. Loss of an X chromosome in a female
· d. Presence of extra X chromosome in male (47 XXY)

Question 21
What is the site of storage and maturation of the sperms of the following?

· a. Efferent ductules
· b. Seminal vesicles
· c. Prostatic gland
· d. Epididymis

Question 22

The medical application of studying Barr bodies is most relevant to which?

· a. Gender determination
· b. Cancer diagnosis
· c. Metabolic disorders
· d. Infectious diseases

Question 23
Regarding to the primordial germ cells?

· a. Give rise to testosterone in the male
· b. Originated in the wall of the yolk sac
· c. Rotated to return their original site
· d. Give rise to estrogen in the female

Question 24
Klinefelter syndrome is associated with which chromosome pattern?

· a. XY
· b. XX
· c. XO
· d. XXY

Question 25

Which of the following does not describe the cell divisions?

· a. Results of meiotic cell division are not identical
· b. Meiotic cell division occurs in male and female germ cells
· c. DNA replication occurs in both types of cell division
· d. Results of mitotic cell division are not identical.

Question 26
Which of the following is the age of puberty in the female?

· a. Period from the second year postnatal to the twelfth year.
· b. Period from 19-21 years postnatal.
· c. Period from 14-16 years postnatal.
· d. Period from 16-18 years postnatal.

Question 27
Which of the following is the definition of conceptus?

· a. Period from the third month to the birth
· b. Period from the third week to the eighth week
· c. Period from labor to end of the first month
· d. Period from fertilization to end of the second week

Question 28
An example of a multinucleated cell is which of these?

· a. Osteoclast
· b. Lymphocyte
· c. Hepatocyte
· d. Plasma cell

Question 29
Nucleoplasm is also known as which?

· a. Karyolymph
· b. Stroma
· c. Cytosol
· d. Hyaloplasm

Question 30
Which of the following is the age of adolescence?

· a. Period from 14-16 years postnatal.
· b. Period from 19-21 years postnatal.
· c. Period from the second year postnatal to the twelfth year.
· d. Period from 16-18 years postnatal.

Question 31
Body of uterus is the site of which of the following?

· a. Implantation
· b. Ovulation
· c. Insemination
· d. Fertilization

Answers to formative 8 “PMS"

---

1. c - Period from the third month to the birth

2. a- Selective for specific substances

3. b- Karyolymph

4. d- Period from fertilization to end of the second week

5. d- Give rise to ovary in the female

6. b- Fertilization

7. d- Osteoclast

8. a- Period from 14-16 years postnatal.

9. b- Chromatin filaments and filamentous RNA

10. c- Dark basophilic membrane

11. a- Epididymis

12. c- Peritoneal cavity

13. c- Period from 16-18 years postnatal.

14. b- XXY

15. a- Period from labor to end of the first month

16. b- Originated in the wall of the yolk sac

17. a- Pars granulosa

18. d- Loss of an X chromosome in a female

19. a- Ovulation

20. a- Period from 19-21 years postnatal.

21. d- Presence of extra X chromosome in male (47 XXY)

22. a- Gender determination

23. d- Two parallel unit membranes

24. a- Period from the third week to the eighth week

25. b- Period from the second year postnatal to the twelfth year.

26. d- Implantation

27. a- Red Blood Cells

28. a- Presence of a third copy of chromosome 21

29. b- Results of mitotic cell division are not identical.

30. c- Sexual maturation

31. c- Insemination

Formative 9 “PMS”

1.

The medical application of studying Barr bodies is most relevant to which?
a. Metabolic disorders
b. Cancer diagnosis
c. Infectious diseases
d. Gender determination

2.

Nucleoplasm is also known as which?
a. Cytosol
b. Karyolymph
c. Stroma
d. Hyaloplasm

3.

Endometrium of the pregnant uterus is called which of the following?
a. Amnion
b. Decidua
c. Yolk sac
d. Allantois

4.

Gonadotropin releasing hormone is secreted by which of the following?
a. Ovary
b. Suprarenal gland
c. Pituitary gland
d. Hypothalamus

5.

Zona pellucida is a glycoprotein secreted by which of the following?
a. Germinal epithelium
b. Theca interna
c. Granulosa cells and oocyte
d. Theca externa

6.

Fertilization occur at which of the following?

a. Intramural part of uterine tube
b. Infundibulum of fallopian tube
c. Isthmus of fallopian tube
d. Ampulla of fallopian tube

7.

The nuclear envelope is formed of which?
a. Continuous membrane without pores
b. A single lipid layer
c. A non-selective barrier
d. Two parallel unit membranes

8.

The exchange through nuclear pores is which of these?
a. Selective for specific substances
b. Only for ions
c. Only for water
d. Non-selective

9.

Where ribosomal assembly within the nucleolus occur?
a. Fibrillar center
b. Pars fibrosa
c. Nuclear pore
d. Pars granulosa

10.

Ovulated mammalian oocyte is arrested at which of the following?
a. Prophase of meiosis I
b. Prophase of meiosis II
c. Metaphase of meiosis I
d. Metaphase of meiosis II

11.

Which of the following best describes the implantation?
a. Completed at the end of 3rd week of pregnancy
b. Begins about day 11 after fertilization
c. Occurs at the embryonic pole
d. The blastocyst is embedded in the myometrium

12.

The blastocyst begins implantation at which of the following?

a. 4–5 days of gestation
b. 2–3 days of gestation
c. 1–2 days of gestation
d. 6–7 days of gestation

13.

The only mammalian cells that are non-nucleated are?
a. Plasma cells
b. Hepatocytes
c. Osteoclasts
d. Red blood cells

14.

The site of storage and maturation of sperms is which of the following?
a. Epididymis
b. Prostate gland
c. Efferent ductules
d. Seminal vesicles

15.

Primary oocyte remains in 1st meiotic division stage till which of the following?
a. Birth
b. Menopause
c. Ovulation
d. 3rd month of intra uterine life

16.

Which of the following best describes spermiogenesis?
a. Secondary spermatocytes undergo mitotic division to form spermatids
b. Transformation of spermatogonia A into spermatocytes
c. Transformation of spermatogonia into spermatids
d. Transformation of spermatids into spermatozoa

17.

Meiosis II of secondary oocytes is completed at which of the following?
a. Ovulation
b. Intrauterine life
c. Before ovulation
d. Fertilization

18.

The part of the sperm responsible for energy production is which of the following?

a. Tail
b. Head
c. Acrosome
d. Midpiece

19.

The pars fibrosa of the nucleolus contains which?
a. Ribosomal subunits
b. Only carbohydrates
c. Dense lipid droplets
d. Chromatin filaments and filamentous RNA

20.

Klinefelter syndrome means which of the following?
a. Deletion of small segment of chromosome 22
b. Loss of an X chromosome in a female
c. Presence of extra X chromosome in male (47 XXY)
d. Presence of a third copy of chromosome 21

21.

In blastocyst, which of the following responsible for formation of the embryo?
a. Trophoblast
b. Inner cell mass
c. Blastocyst cavity
d. Zona pellucida

22.

Follicle stimulating hormone is secreted by which of the following?
a. Suprarenal gland
b. Pituitary gland
c. Hypothalamus
d. Ovary

23.

Which of the following best describes capacitation of the sperm?
a. Removal of glycoprotein coat from the sperm head
b. Caused by the zona pellucida
c. Removal of the tail
d. Prevents polyspermy

24.

An example of a multinucleated cell is which of these?

a. Lymphocyte
b. Hepatocyte
c. Osteoclast
d. Plasma cell

25.

Which of the following does not describe the cell divisions?
a. Results of meiotic cell division are not identical
b. Meiotic cell division occurs in male and female germ cells
c. Results of mitotic cell division are not identical
d. DNA replication occurs in both types of cell division

26.

Which of the following occurs in mid-cycle by LH secretion?
a. Causes follicular rupture and ovulation
b. Result from stimulation of anterior pituitary by estrogen
c. Stimulate oocyte to complete Meiosis I and enter Meiosis II
d. Stimulate production of gonadotrophins by follicular stroma cells

27.

Normal site of implantation is which of the following?
a. Uterus
b. Ovary
c. Fallopian tube
d. Vagina

28.

The nuclear envelope is visible under a light microscope as a?
a. Series of granules
b. Light eosinophilic line
c. Dark basophilic membrane
d. Clear space

29.

Turner syndrome means which of the following?
a. Deletion of small segment of chromosome 22
b. Presence of a third copy of chromosome 21
c. Loss of an X chromosome in a female
d. Presence of extra X chromosome in male (47 XXY)

30.

The most important region of the decidua for the nourishment of conceptus is which of the following?
a. Decidua basalis
b. Decidua capsularis
c. Chorion frondosum
d. Decidua parietalis

31.

Down syndrome means which of the following?
a. Deletion of small segment of chromosome 22
b. Presence of a third copy of chromosome 21
c. Presence of extra X chromosome in male (47 XXY)
d. Loss of an X chromosome in a female

32.

Haploid nuclei that fuse at fertilization are called which of the following?
a. Morula
b. Pronuclei
c. Zygote
d. Blastomeres

33.

Klinefelter syndrome is associated with which chromosome pattern?
a. XX
b. XXY
c. XY
d. XO

34.

Luteinizing hormone is secreted by which of the following?
a. Suprarenal gland
b. Hypothalamus
c. Ovary
d. Pituitary gland

ANSWERS

1. d
2. b

3. b
4. d
5. c
6. d
7. d
8. a
9. d
10. a
11. c
12. d
13. d
14. a
15. c
16. d
17. d
18. d
19. d
20. c
21. b
22. b
23. a
24. c
25. c
26. a
27. a
28. c
29. c
30. a
31. b
32. b
33. b
34. d

Formative 10 “PMS”

Question 1

Which of the following occurs in the 2nd week of pregnancy?
a. Embryonic folding
b. Formation of primitive streak
c. Beginning of gastrulation
d. Formation of bilaminar germ disc

Question 2

Embryoblast gives which of the following in the second Week of development?
a. Cytotrophoblast and syncytotrophoblast
b. Epiblast and hypoblast
c. Ectoderm and mesoderm
d. Amniotic cavity and yolk sac

Question 3

Which statement of the following best describes the primordial germ cells?
a. Give rise to ova in the female
b. Give rise to testes in the male
c. Originated in the wall of the amniotic cavity
d. Ectodermal in origin

Question 4

What is the site of implantation of the blastocyst?
a. Body of the uterus
b. Uterine tube
c. Douglas pouch
d. Ovary

Question 5

Myoepithelium is an example of epithelial tissue specialized for which of these functions?
a. Filtration
b. Secretion
c. Contractile function
d. Sensory reception

Question 6

Intermediate mesoderm will give rise to which of the following?
a. Heart
b. Somites
c. Neural tube
d. Kidneys and gonads

Question 7

The basal lamina is described as PAS+ve, which indicates the presence of?
a. Lipids
b. Nucleic acids
c. Keratin
d. Glycoproteins

Question 8

Neural tube will give rise to which of the following?
a. Kidneys and gonads
b. Neural tube
c. Central nervous system
d. Heart

Question 9

Which of the following is responsible for formation of the embryo?
a. Zona pellucida
b. Blastocyst cavity
c. Trophoblast

Question 10

Which intercellular junction is the most apical in epithelial cells?
a. Tight junctions
b. Gap junctions
c. Zonulae adherens
d. Desmosomes

Question 11

Desmosomes link to which cytoskeletal component via desmoplakin?
a. Actin filaments
b. Myosin filaments
c. Intermediate filaments
d. Microtubules

Question 12

Epithelial cells receive their nutrients from which of these?
a. The apical surface via diffusion from the lumen
b. Specialized secretory ducts
c. The underlying connective tissue
d. Direct blood supply within the epithelium

Question 13

Which of the following occurs in the 3rd week of pregnancy?
a. Formation of amniotic cavity
b. Formation of bilaminar germ disc
c. Formation of yolk sac
d. Beginning of gastrulation

Question 14

Which of the following are developed from endoderm?
a. Mucous membranes of the gastrointestinal tract
b. Kidneys
c. Dermis of the skin
d. Epidermis of the skin

Question 15

Which of the following statements does not describe the cleavage?
a. It occurs while the zygote is passing through the fallopian tube
b. Its cells are called blastomeres
c. It is a series of mitotic divisions
d. Results in an increase in the size of the cells in the zygote

---

Question 16

Cilia are anchored at the base by which of these?
a. Zonula adherens
b. Basal body
c. Hemidesmosome
d. Terminal web

Question 17

Which one of the following statements describes the implantation?
a. Begins about day 11 after fertilization
b. Completed at the end of 3rd week of pregnancy
c. Occurs at the embryonic pole
d. The blastocyst is embedded in the myometrium

Question 18

Microvilli are mainly found in epithelia specialized for which of these?
a. Secretion
b. Contraction
c. Absorption
d. Protection

Question 19

Which of the following means gastrulation?
a. Formation of bilaminar germ disc
b. Formation of trilaminar germ disc
c. Embryonic folding
d. Formation of primitive streak

Question 20

Which statement best describes capacitation of the sperm?
a. Prevents polyspermy
b. Caused by the zona pellucida
c. Means removal of the tail
d. Means removal of glycoprotein coat from the sperm head

Question 21

Small invaginations projecting from the connective tissue into the epithelium, which increase the area of contact, are called?
a. Basal laminae
b. Hemidesmosomes
c. Cellular junctions
d. Papillae

Question 22

Which of the following are developed from mesoderm?
a. Muscles of the limbs
b. Mucous membranes of the respiratory tract
c. Mucous membranes of the gastrointestinal tract
d. Mucous membranes of part of urinary bladder

Question 23

In desmosomes, transmembrane proteins belong to which family?
a. Integrins
b. Adherins
c. Claudins
d. Occludins

Question 24

Which one of the following statements describes the implantation?
a. Occurs at the abembryonic pole
b. The blastocyst is embedded in the basal layer of the endometrium
c. Completed at the end of 3rd week of pregnancy
d. Begins about day 7 after fertilization

Question 25

Which of the following are developed from notochord?
a. Apical ligament
b. Muscles of the limbs
c. Mucous membranes of the respiratory tract
d. Mucous membranes of part of urinary bladder

Question 26

Which one of the following statements describes the implantation?
a. Completed at the end of 3rd week of pregnancy
b. Begins about day 11 after fertilization
c. Occurs at the abembryonic pole
d. The blastocyst is embedded in superficial layers of the endometrium

Question 27

Trophoblast gives which of the following in the second Week of development?
a. Amniotic cavity and yolk sac
b. Epiblast and hypoblast
c. Cytotrophoblast and syncytiotrophoblast
d. Ectoderm and mesoderm

Question 28

What cytoskeletal filament is abundant in microvilli?
a. Intermediate filaments
b. Actin filaments
c. Myosin filaments
d. Microtubules

Question 29

Which of the following is an ectodermal structure?
a. Cartilage and bone
b. Medulla of the suprarenal gland

c. Cortex of the suprarenal gland
d. Cardiovascular system

Question 30

Which of the following are developed from notochord?
a. Brain
b. Spinal cord
c. Nucleus pulposus of intervertebral disc
d. Annulus fibrosus of intervertebral disc

Question 31

Which statement best describes the primordial germ cells?
a. Give rise to ovary in the female
b. Originated in the wall of the amniotic cavity
c. Ectodermal in origin
d. Give rise to sperms in the male

Question 32

Which one of the following statements describes the implantation?
a. Completed at the end of 2nd week of pregnancy
b. Occurs at the abembryonic pole
c. Begins about day 11 after fertilization
d. The blastocyst is embedded in superficial layers of the endometrium

ANSWERS

1. d
2. b
3. a
4. a
5. c
6. d

7. d
8. c
9. a (Inner cell mass)
10. a
11. c
12. c
13. d
14. a
15. d
16. b
17. c
18. c
19. b
20. d
21. d
22. a
23. b
24. d
25. a
26. d
27. c
28. b
29. b
30. c
31. d
32. a

Formative 11 “PMS” with answers

Prepared by group “A”

---

1. At what age of head of the fetus is about one fourth in relation to rest of the body?

a. The 5th month
b. The 3rd month
c. The 2nd month
d. At birth

___

2. Which one of the following is formed during the second month of development?

a. Joints
b. Cartilages
c. Eyes
d. Bones

___

3. Small invagination projecting from the connective tissue into the epithelium, which increase the area of contact, are called?

a. Cellular junctions
b. Papillae
c. Basal laminae
d. Hemidesmosomes

___

4. Which one of the following is a main external feature at the end of the fourth Week?

a. Pharyngeal arches
b. Joints
c. Cartilages
d. Bones

___
5. Which one of the following is formed during the second month of development?

a. Face
b. Cartilages
c. Bones
d. Joints

___
6. At what age does the embryo has a distinct human appearance?

a. 32 days
b. 56 days
c. 42 days
d. 28 days

___
7. At what age of head of the fetus is about one half in relation to rest of the body?

a. The 3rd month
b. At birth
c. The 2nd month
d. The 5th month

___
8. At what age of the following all the organs and limbs are present?

a. End of the 4th month
b. End of the 3rd month
c. End of the 2nd month
d. End of the 5th month

___
9. What cytoskeletal filament is abundant in microvilli?

a. Myosin filaments
b. Intermediate filaments
c. Actin filaments
d. Microtubules

___
10. Which of the following is a primordium of the umbilical cord?

a. Extraembryonic coelom
b. Cytotrophoblast
c. Allantois
d. Connecting stalk

___
11. Myoepithelium is an example of epithelial tissue specialized for which function?

a. Contractile function
b. Secretion
c. Sensory reception
d. Filtration

___
12. Which intercellular junction is the most apical in epithelial cells?

a. Desmosomes
b. Gap junctions
c. Zonulae adherens
d. Tight junctions

___
13. What is the CRL of the embryo in the fifth Week?

a. 28–30 mm
b. 17–22 mm
c. 5–8 mm
d. 10–14 mm

___

14. Vitelline duct connects definitive yolk sac with which of the following?

a. Hindgut
b. Foregut
c. Amnion
d. Midgut

___

15. At what age of head of the fetus is about one third in relation to rest of the body?

a. At birth
b. The 5th month
c. The 2nd month
d. The 3rd month

___

16. Desmosomes link to which cytoskeletal component via desmoplakin?

a. Actin filaments
b. Microtubules
c. Myosin filaments
d. Intermediate filaments

___

17. Epithelial cells receive their nutrients from which of these?

a. The underlying connective tissue
b. Specialized secretory ducts
c. The apical surface via diffusion from the lumen
d. Direct blood supply within the epithelium

___

18. Which one of the following is a main external feature at the end of the fourth Week?

a. Bones
b. Cartilages
c. Joints
d. Somites

___

19. The basal lamina is described as PAS+ve, which indicates the presence of?

a. Nucleic acids
b. Glycoproteins
c. Keratin
d. Lipids

___

20. Microvilli are mainly found in epithelia specialized for which of these?

a. Protection
b. Secretion
c. Contraction
d. Absorption

___

21. Which of the following is an anomaly in the position of the placenta?

a. Succenturiate placenta
b. Placenta tripartite
c. Placenta brevia
d. Membranous placenta

___

22. Which of the following is derived from endoderm?

a. Anterior lobe of pituitary gland
b. Nervous tissue
c. Lens
d. Epithelium of upper half of anal canal

___

23. Which one of the following is formed during the second month of development?

a. Limbs
b. Cartilages
c. Joints
d. Bones

___

24. When the baby will be designated male or female at birth?

a. 2nd month
b. 8th month
c. 6th month
d. 4th month

___
25. Which of the following is a neural crest derivative?

a. The cardiovascular system
b. Dorsal root ganglia of the spinal nerves
c. Cartilage and bone
d. Cortex of the suprarenal gland

___
26. Which of the following are developed from lateral plate mesoderm?

a. Mucus membrane of part of urinary bladder
b. Muscles of the limbs
c. Mucous membrane of the gastro-intestinal tract
d. Mucus membrane of the respiratory tract

___
27. In desmosomes, transmembrane proteins belong to which family?

a. Cadherins
b. Claudins
c. Occludins
d. Integrins

___
28. Cilia are anchored at the base by which of these?

a. Hemidesmosome
b. Terminal web
c. Zonula adherens
d. Basal body

___
29. Which one of the following is formed during the second month of development?

a. Bones
b. Nose
c. Cartilages
d. Limbs

___
30. Which one of the following is a derivative of the ectoderm?

a. Rectum
b. Brain
c. Cartilage
d. Bones

Answers to formative 11 “PMS”

---

1. c. The 2nd month

2. b. Cartilages

3. b. Papillae

4. a. Pharyngeal arches

5. a. Face

6. b. 56 days

7. c. The 2nd month

8. c. End of the 2nd month

9. c. Actin filaments

10. d. Connecting stalk

11. a. Contractile function

12. d. Tight junctions

13. c. 5–8 mm

14. d. Midgut

15. d. The 3rd month

16. d. Intermediate filaments

17. a. The underlying connective tissue

18. d. Somites

19. b. Glycoproteins

20. d. Absorption

21. c. Placenta brevia

22. d. Epithelium of upper half of anal canal

23. c. Joints

24. d. 4th month

25. b. Dorsal root ganglia of the spinal nerves

26. b. Muscles of the limbs

27. a. Cadherins

28. d. Basal body

29. b. Nose

30. b. Brain

Formative 12 “PMS” with answers

Prepared by group “A”

---

1. Which statement of the following best describes fetal functions of amnion?

a. Forms adhesions between different parts of the embryo.
b. Allows fetal movements.
c. Excretion.
d. Protection against internal trauma.

---

2. Which statement of the following best describes the chorion?

a. Primary villi; formed of embryoblast.
b. Chorion leave: results from regeneration of villi
c. Secondary villi; formed of trophoblast and intraembryonic mesoderm
d. Membrane formed of trophoblast and its lining mesoderm

---

3. Which of the following is a congenital anomaly of the vitelline duct?

a. Uracheal fistula.
b. Mickel’s Diverticulum.
c. Uracheal cyst.
d. Uracheal sinus.

---

4. Which of the following best describes the fetal membranes?

a. Connective tissue of the umbilical cord is called urachus
b. Umbilical cord connects the fetus to the myometrium of the uterus
c. Allantois is a diverticulum from the caudal part of the yolk sac
d. Umbilical cord has two veins and one artery

---

5. Which of the following best describes the following embryonic structure?

a. Cytotrophoblast is inner layer of stratified cells
b. Syncytiotrophoblast is outer multinucleated zone
c. Hypoblast layer is formed of columnar cells
d. Amniotic cavity appears within hypoblast

---

6. Type III collagen is the main constituent of which?

a. Lamina densa
b. Elastic fibers
c. Dense regular connective tissue
d. Reticular fibers

---

7. Which of the following will be changed into ligamentum teres of the liver?

a. Closure of the umbilical arteries
b. Closure of the left umbilical vein
c. Closure of the ductus arteriosus
d. Closure of the urachus

---

8. Intermediate mesoderm will give rise to which of the following?

a. Kidneys
b. Neural Tube
c. Somites
d. Heart

---

9. The characteristic "cart-wheel" or "clock-face" nuclear chromatin pattern is seen in which type of cells?

a. Fibroblasts
b. Mast cells
c. Plasma cells
d. Macrophages

---

10. What is the primary function of plasma cells?

a. Phagocytosis
b. Synthesis and secretion of antibodies
c. Synthesis of tropocollagen
d. Storage of heparin and histamine

---

11. Mast cell granules stain metachromatically and contain which?

a. Fat droplets
b. Heparin and histamine
c. Antibodies and interferon
d. Collagen and elastin

---

12. Which of the following best describes folding?

a. Brain forms most caudal part of the embryo
b. Embryo acquires shape
c. Heart becomes dorsal in position
d. Septum transversarum becomes lateral in position

---

13. Which of the following structures is not a content of full-term umbilical cord?

a. Left umbilical vein
b. Right umbilical vein
c. Umbilical arteries
d. Wharton’s jelly

---

14. Which of the following is the maternal part of the placenta?

a. Chorion frondosum
b. Decidua parietalis
c. Decidua basalis
d. Decidua capsularis

---

15. What is the normal amount of amniotic fluid?

a. 6 liters
b. Three liters
c. One liter
d. Half a liter

---

16. Which of the followings is a function of the amnion?

a. Show
b. Adhesions
c. Hydramnios
d. Oligohydramnios

---

17. Which vital stain is used to identify macrophages?

a. Orcein
b. Silver stain
c. H&E
d. Trypan blue

---

18. Which of the following are developed from lateral plate mesoderm?

a. Mucous membrane of the gastro-intestinal tract
b. Mucus membrane of part of urinary bladder
c. Muscles of the limbs
d. Mucus membrane of the respiratory tract

---

19. Plasma cells are derived from which cell?

a. B-lymphocytes
b. Monocytes
c. T-lymphocytes
d. Fibroblasts

---

20. Type II collagen is a major component of which?

a. Basal lamina
b. Hyaline and elastic cartilage
c. Skin tendons.
d. Bone

---

21. Paraxial mesoderm will give rise to which of the following?

a. Neural tube
b. Gonads
c. Heart
d. Somites

---

22. Reticular fibers are best demonstrated with which stain?

a. H&E stain
b. Silver stains
c. Vital stains like Trypan blue
d. Orcein stain

---

23. Which of the following best describes the location of placenta previa?

a. At surface of the ovary
b. Close to the internal os of the cervix
c. In peritoneal cavity
d. In the uterine tube

---

24. Which statement of the following best describes the primordial germ cells?

a. Rotate to reach their terminal site
b. Originated in the wall of the yolk sac
c. Give rise to testis in the male
d. Give rise to ovary in the female

—

25. Vitelline duct connects definitive yolk sac with which of the following?

a. Amnion
b. Hindgut
c. Midgut
d. Foregut

---

26. Type I collagen is predominantly found in which of these?

a. Basement membranes
b. Reticular fibers
c. Hyaline cartilage
d. Bone

---

27. Which statement of the following best describes the chorion frondosum?

a. Lies opposite decidua parietalis as pregnancy advances
b. The part of chorion that does not participate in exchange process
c. Decreases in size and complexity on the embryonic pole
d. Fetal part of the placenta

---

28. Which statement of the following best describes the too long umbilical cord?

a. May encircle body of the placenta
b. May decrease to 20 cm
c. May form knots which increase the blood supply to the fetus
d. May prolapse before delivery of the child

---

29. Which of the following is a primordium of the umbilical cord?

a. Extraembryonic coelom
b. Allantois
c. Cytotrophoblast
d. Connecting stalk

---

30. Which one of the following is a fetal membrane?

a. Notochord
b. Folding
c. Decidua
d. Amnion

---

31. The "signet-ring" appearance in H&E stained sections is characteristic of which cell type?

a. Plasma cell
b. Fat cell
c. Mast cell
d. Macrophage

---

32. Which of the following will be changed into the median umbilical ligament?

a. Closure of the umbilical arteries
b. Closure of the ductus arteriosus
c. Closure of the left umbilical vein
d. Closure of the urachus

Answers to formative 12 “PMS”

1. b. Allows fetal movements

2. d. Membrane formed of trophoblast and its lining mesoderm

3. b. Mickel’s Diverticulum

4. c. Allantois is a diverticulum from the caudal part of the yolk sac

5. b. Syncytiotrophoblast is outer multinucleated zone

6. d. Reticular fibers

7. b. Closure of the left umbilical vein

8. a. Kidneys

9. c. Plasma cells

10. b. Synthesis and secretion of antibodies

11. b. Heparin and histamine

12. b. Embryo acquires shape

13. b. Right umbilical vein

14. c. Decidua basalis

15. c. One liter

16. a. Show

17. d. Trypan blue

18. c. Muscles of the limbs

19. a. B-lymphocytes

20. b. Hyaline and elastic cartilage

21. d. Somites

22. b. Silver stains

23. b. Close to the internal os of the cervix

24. b. Originated in the wall of the yolk sac

25. c. Midgut

26. d. Bone

27. d. Fetal part of the placenta

28. d. May prolapse before delivery of the child

29. d. Connecting stalk

30. d. Amnion

31. b. Fat cell

32. d. Closure of the urachus

Formative 13 “PMS” with answers

Prepared by group “A”

---

1. What is the normal amount of the amniotic fluid?
    a. Three liters
    b. Half a liter
    c. 6 liters
    d. One liter

****

2. A structure situated nearer to the median plane than another is which of following?
    a. Distal
    b. Superior
    c. Medial
    d. Lateral

****

3. Reticular fibers are best demonstrated with which stain?
    a. Vital stains like Trypan blue
    b. Silver stains
    c. Orcein stain
    d. H&E stain

****

4. The "signet-ring" appearance in H&E stained sections is characteristic of which cell type?
    a. Macrophage
    b. Mast cell
    c. Fat cell
    d. Plasma cell

****

5. What is the fetal part of the placenta?
   a. Decidua basalis
   b. Decidua capsularis
   c. Chorion frondosum
   d. Chorion leave

****

6. Sutures of the skull is an example of which type of the joint?
   a. Synovial
   b. Secondary cartilaginous
   c. Primary cartilaginous
   d. Fibrous

****

7. Which of the following is an anomaly in the position of the placenta?
   a. Placenta previa
   b. Membranous placenta
   c. Succenturiate placenta
   d. Placenta tripartite

****

8. What is the primary function of plasma cells?
   a. Synthesis and secretion of antibodies
   b. Storage of heparin and histamine
   c. Phagocytosis
   d. Synthesis of tropocollagen

****

9. Which of the following is considered as a secondary cartilaginous joint?
   a. Junction between epiphysis and diaphysis
   b. Symphysis pubis
   c. Shoulder
   d. Suture

****

10. Type II collagen is a major component of which?
   a. Basal lamina
   b. Bone
   c. Skin tendons
   d. Hyaline and elastic cartilage

****

11. The characteristic "cart-wheel" or "clock-face" nuclear chromatin pattern is seen in which type of cells?

a. Plasma cells
b. Mast cells
c. Fibroblasts
d. Macrophages

****

12. Type I collagen is predominantly found in which of these?

a. Bone
b. Reticular fibers
c. Hyaline cartilage
d. Basement membranes

****

13. Which of the following is a part of the digestive system?

a. Esophagus
b. Trachea
c. Lung
d. Nose

****

14. Which of the following is a fetal function of amnion during labor?

a. Pushes out the water which closes the cervical canal
b. Forms a bag of mucous
c. Helps the dilatation of the cervix of the uterus
d. Acts as a natural septic of the birth canal

****

15. Pneumatic bones are present in which of the following?

a. Femur
b. Skull
c. Humerus
d. Radius

****

16. Plasma cells are derived from which cell?

a. B-lymphocytes
b. T-lymphocytes
c. Monocytes
d. Fibroblasts

****

17. What is the type of elbow joint?
    a. Hinge synovial joint
    b. Ball and socket synovial joint
    c. Plane synovial joint
    d. Pivot synovial joint

****

18. Which is the statement best describes monozygotic twins of the following?
    a. Occurs when one fertilized ovum splits into two or more embryos
    b. Splitting at two cell stage; two embryos will have shared placenta
    c. May be of the same or different sex
    d. Splitting at two cell stage; two embryos will have shared chorionic sac

****

19. Which of the following is the site of fertilization?
    a. Infundibulum of uterine tube
    b. Ampulla of uterine tube
    c. Isthmus of uterine tube
    d. Upper part of uterine cavity

****

20. Which of the following is a primordium of the umbilical cord?
    a. Extraembryonic coelom
    b. Cytotrophoblast
    c. Allantois
    d. Connecting stalk

****

21. The common bile duct opens into which of the followings?
    a. Third part of the duodenum
    b. Second part of the duodenum
    c. First part of the duodenum
    d. Fourth part of the duodenum

****

22. Which of the following is an anatomical end artery?
    a. Arteries of the stomach
    b. Coronary arteries
    c. Central artery of the retina
    d. Genicular arteries

****

23. The heart distributes blood to the whole body through which of the followings?

a. Pulmonary veins
b. Aorta
c. Coronary arteries
d. Pulmonary trunk

****

24. Which of the followwing is the type of the vertebrae according to the shape?

a. Flat
b. Irregular
c. Long
d. Short

****

25. Mast cell granules stain metachromatically and contain which?

a. Collagen and elastin
b. Fat droplets
c. Antibodies and interferon
d. Heparin and histamine

****

26. Teratology is the science that studies which of the following?

a. Normal development during the third trimester
b. Normal development during the first trimester
c. Birth defects
d. Normal development during the second trimester

****

27. Gastrulation of the embryo occurs at which of the following weeks?

a. Second week
b. First week
c. Fourth week
d. Third week

****

28. What is the second envelope of the body?

a. Superficial fascia.
b. Deep fascia.
c. Skin.
d. Muscles

****

29. Which vital stain is used to identify macrophages?
    a. Orcein
    b. Silver stain
    c. Trypan blue
    d. H&E

****

30. Which one of the following is an association?
    a. CHARGE
    b. Klinefelter's
    c. Down
    d. Turner

****

31. Which of the followings is a congenital anomaly of the vitelline duct?
    a. Uracheal cyst
    b. Uracheal fistula
    c. Mickel's diverticulum
    d. Uracheal sinus

****

32. Which one of the following is a constituent of the placenta?
    a. Umbilical arteries
    b. Chorionic blood vessels
    c. Warton's jelly
    d. Allantois

****

33. Which of the following is a part of the respiratory system?
    a. Stomach
    b. Trachea
    c. Kidney
    d. Esophagus

****

34. Type III collagen is the main constituent of which?
    a. Reticular fibers
    b. Dense regular connective tissue
    c. Lamina densa
    d. Elastic fibers

****

35. Ribs and sternum are parts of which of the following?
    a. Axial skeleton
    b. Appendicular skeleton
    c. Bone of the upper limb
    d. Bones of the lower limb

Answers to formative 13 “PMS”

---

1. d. One liter
2. c. Medial
3. b. Silver stains
4. c. Fat cell
5. c. Chorion frondosum
6. d. Fibrous
7. a. Placenta previa
8. a. Synthesis and secretion of antibodies
9. b. Symphysis pubis

10. d. Hyaline and elastic cartilage

11. a. Plasma cells

12. a. Bone

13. a. Esophagus

14. c. Helps the dilatation of the cervix of the uterus

15. b. Skull

16. a. B-lymphocytes

17. a. Hinge synovial joint

18. a. Occurs when one fertilized ovum splits into two or more embryos

19. b. Ampulla of uterine tube

20. d. Connecting stalk

21. b. Second part of the duodenum

22. c. Central artery of the retina

23. b. Aorta

24. b. Irregular

25. d. Heparin and histamine

26. c. Birth defects

27. d. Third week

28. a. Superficial fascia

29. c. Trypan blue

30. a. CHARGE

31. c. Mickel’s diverticulum

32. b. Chorionic blood vessels

33. b. Trachea

34. a. Reticular fibers

35. a. Axial skeleton

Formative 14 “PMS” with answers.

Prepared by group “A”

---

1. What is the type of collagen fiber in bone?
    a. Type I.
    b. Type III.
    c. Type IV.
    d. Type II.

****

2. What is the stabilizing cell structure that resists tension?
    a. Intermediate filaments
    b. Microfilaments
    c. Microtubule
    d. Myosin filaments

****

3. What is the type of wrist joint?
    a. Plane synovial joint
    b. Pivot synovial joint
    c. Ball and socket synovial joint
    d. Ellipsoid synovial joint

****

4. Which of the following occurs in the 2nd week of pregnancy?
    a. Formation of bilaminar germ disc
    b. Formation of the three germ layers
    c. Formation of the neural tube
    d. Embryonic folding

****

5. Which statement of the following best describes the chorion frondosum?

a. The part of chorion does not participate in exchange process
b. Lies opposite decidua parietalis as pregnancy advance
c. Fetal part of placenta
d. Increase in size and complexity on the abembryonic pole

****

6. Which cell is increased in parasitic diseases?

a. Microphage
b. Basophil
c. Eosinophil
d. Neutrophil

****

7. What is the part of the sperm responsible for energy production?

a. Midpiece
b. Acrosome
c. Tail
d. Head

****

8. What is the distinguishing character of the hyaline cartilage matrix?

a. Contains collagen type I
b. Calcified matrix
c. Contains collagen type II
d. Vascular

****

9. What is the lining of the absorptive membranes?

a. Ciliated cells
b. Columnar cells with microvilli
c. Simple squamous epithelium
d. Cells containing numerous secretory granules

****

10. Which is the bone forming cell?

a. Osteocyte
b. Osteogenic
c. Osteoblast
d. Osteoclast

****

11. What is the distinguishing character of epithelial tissues?
    a. Numerous blood vessels
    b. Abundant extracellular matrix
    c. Widely separated cells
    d. Closely aggregated cells

****

12. The heart distributes blood to the whole body through which of the following?
    a. Pulmonary veins
    b. Coronary arteries
    c. Pulmonary trunk
    d. Aorta

****

13. Which cell synthesizes tropocollagen?
    a. Fibrocyte
    b. Fat cell
    c. Plasma cells
    d. Fibroblasts

****

14. Which of the following are developed from endoderm?
    a. Epidermis of the skin
    b. Kidneys
    c. Mucous membranes of the gastrointestinal tract
    d. Dermis of the skin

****

15. What is the site of storage and maturation of sperms?
    a. Epididymis
    b. Prostatic gland
    c. Seminal vesicles
    d. Efferent ductules

****

16. Which of the following occurs in the 3rd week of pregnancy?
    a. Formation of bilaminar germ disc
    b. Beginning of gastrulation
    c. Formation of amniotic cavity
    d. Formation of yolk sac

****

17. Which of the following statements does not describe cleavage?
    a. Results in an increase in the size of the cells
    b. Occurs while passing through the fallopian tube
    c. Series of meiotic divisions
    d. Its cells are called somitomers

****

18. Where is the osteon present?
    a. Cancellous bone
    b. Compact bone
    c. Spongy bone
    d. Woven bone

****

19. In which cell are ribosomes absent?
    a. Lymphocytes
    b. Kidney cells
    c. Muscle cells
    d. Erythrocytes

****

20.Primary oocyte remains in 1st meiotic division till?
    a. 3rd month of intrauterine life
    b. Before ovulation
    c. Birth
    d. Menopause

****

21. Type of collagen fiber in compact bone is:
    a. Type I
    b. Type II
    c. Type VII
    d. Type III

****

22. Which statement best describes primordial germ cells?
    a. Give rise to testes in the male
    b. Give rise to ova in the female
    c. Ectodermal in origin
    d. Originated in the wall of the amniotic cavity

****

23. What is the length of each uterine tube?
    a. 15 cm
    b. 5 cm
    c. 10 cm
    d. 20 cm

****

24. What is the distinguishing character of hemidesmosomes?
    a. Present between cells and basement membrane
    b. Present between adjacent epithelial cells
    c. Help rapid nerve conduction
    d. Seal adjacent cells

****

25. Which statement best describes kidney anatomy?
    a. Adult kidneys lie in pelvis
    b. Ureter conducts urine to bladder
    c. Suprarenal gland covers lower pole
    d. Left kidney lower than right

****

26. Which is responsible for formation of trophoblast?
    a. Inner cell mass
    b. Zona pellucida
    c. Outer cell mass
    d. Blastocyst cavity

****

27. Transitional epithelium lines:
    a. Trachea
    b. Cornea
    c. Ureter
    d. Esophagus

****

28. What is the dividing cell in appositional growth?
    a. Chondroblast
    b. Osteocyte
    c. Chondrogenic cell
    d. Chondrocyte

****

29. Which is part of the respiratory system?
    a. Stomach
    b. Esophagus
    c. Lung
    d. Kidney

****

30. Which statement describes implantation?
    a. Embedded in superficial endometrium
    b. Begins day 11
    c. Occurs at abembryonic pole
    d. Completed end of 3rd week

****

31. Distinguishing character of brown fat cells:
    a. Pigmented fat
    b. Numerous in adult
    c. Unilocular
    d. Multilocular

****

32. Primordial germ cells:
    a. Ectodermal in origin
    b. Give rise to ovary
    c. Give rise to sperms
    d. Originated in wall of amnion

****

33. Developed from notochord:
    a. Nucleus pulposus
    b. Annulus fibrosus
    c. Brain
    d. Spinal cord

****

34. Neural tube gives rise to:
    a. CNS
    b. Muscles
    c. Heart
    d. Kidneys and gonads

****

35. Does not describe fertilization:
    a. Ovulation
    b. Formation of sperms
    c. Shedding cytoplasm
    d. Capacitation

****

36. Meiosis II completed at:
    a. Fertilization
    b. Intrauterine life
    c. Before ovulation
    d. Ovulation

****

37. Structure of nuclear envelope:
    a. Unit membrane with cristae
    b. Acidophilic membrane
    c. Unit membrane with ribosomes
    d. Double unit membrane

****

38. Primitive streak appears in:
    a. First week
    b. Fourth week
    c. Second week
    d. Third week

****

39. Metachromatic granules present in:
    a. Pigment cells
    b. Fat cells
    c. Plasma cells
    d. Mast cells

****

40. Cartilage forming fetal skeleton:
    a. Elastic
    b. Fibrocartilage
    c. Yellow
    d. Hyaline

****

41. Fertilization occurs in:
    a. Infundibulum
    b. Intramural part
    c. Ampulla
    d. Isthmus

****

42. Implantation:
    a. Completed end of 3rd week
    b. Begins day 11
    c. Occurs at embryonic pole
    d. Embedded in myometrium

****

43. Endometrium of pregnant uterus:
    a. Decidua
    b. Chorion
    c. Umbilical cord
    d. Amnion

****

44. Derived from neural crest:
    a. Cartilage and bone
    b. Medulla of suprarenal gland
    c. Cortex of suprarenal gland
    d. Cardiovascular system

****

45. Gastrulation means:
    a. Formation of trilaminar disc
    b. Embryonic folding
    c. Formation of primitive streak
    d. Bilaminar disc

****

46. Secondary cartilaginous joint:
    a. Suture
    b. Manubriosternal joint
    c. Shoulder
    d. Epiphysis–diaphysis junction

****

47. Structure of microvilli:
    a. Microfilaments
    b. Microtubules
    c. Intermediate filaments
    d. Thick filaments

****

48. Occurs in 2nd week:
    a. Primitive streak
    b. Bilaminar germ disc
    c. Gastrulation
    d. Folding

****

49. Bone growth defect in short stature:
    a. Interstitial
    b. Appositional
    c. Epiphyseal
    d. Both

****

50. Shape of neutrophil nucleus:
    a. Irregular
    b. Rounded
    c. Multilobed
    d. Kidney-shaped

****

51. Developed from notochord:
    a. Respiratory mucosa
    b. Limb muscles
    c. Apical ligament
    d. Part of urinary bladder mucosa

****

52. Does not describe spermatogenesis:
    a. Begins at birth
    b. Occurs in testis
    c. Ends at adult age
    d. Primary spermatocytes divide mitotically

****

53. Capacitation means:
    a. Removal of glycoprotein coat
    b. Prevents polyspermy
    c. Caused by zona pellucida
    d. Removal of tail

****

54. Intermediate mesoderm gives rise to:
    a. Neural tube
    b. Kidneys and gonads
    c. Heart
    d. Somites

****

55. Congenital anomaly of amnion:
    a. Adhesions
    b. Forms sac
    c. Allows movement
    d. Protection

****

56. Developed from mesoderm:
    a. GIT mucosa
    b. Limb muscles
    c. Respiratory mucosa
    d. Urinary bladder mucosa

****

57. Gomphosis is:
    a. Primary cartilaginous
    b. Synovial
    c. Fibrous
    d. Secondary cartilaginous

****

58. Progenitor cells:
    a. Daughter cells of pluripotent stem cells
    b. Stem cells
    c. Slowly dividing
    d. Divide infrequently

****

59. Reticular fibers:
    a. Dark brown fresh
    b. Visible by H&E
    c. Yellowish fresh
    d. Dark brown by silver stain

****

60. Multinucleated cell:
    a. Osteocyte
    b. Osteoclast
    c. Macrophage
    d. Osteoblast

****

61. Main pancreatic duct opens into:
    a. 4th part of duodenum
    b. 1st part
    c. 2nd part
    d. 3rd part

****

62. Part of digestive system:
    a. Stomach
    b. Trachea
    c. Nose
    d. Lung

****

63. Second envelope of body:
    a. Skin
    b. Fascia
    c. Deep fascia
    d. Muscles

****

64. Blastocyst begins implantation at:
    a. 6th day
    b. 3rd day
    c. 10th day
    d. 8th day

****

Answers to formative 14 “PMS”

1. a – Type I

2. a – Intermediate filaments

3. d – Ellipsoid synovial joint

4. a – Formation of bilaminar germ disc

5. c – Fetal part of placenta

6. c – Eosinophil

7. a – Midpiece

8. c – Contains collagen type II

9. b – Columnar cells with microvilli

10. c – Osteoblast

11. d – Closely aggregated cells

12. d – Aorta

13. d – Fibroblasts

14. c – Mucous membranes of the gastrointestinal tract

15. a – Epididymis

16. b – Beginning of gastrulation

17. c – Series of meiotic divisions

18. b – Compact bone

19. d – Erythrocytes

20. c – Birth

21. a – Type I

22. b – Give rise to ova in the female

23. c – 10 cm

24. a – Present between cells and basement membrane

25. b – The ureter conducts urine to the bladder

26. c – Outer cell mass

27. c – Ureter

28. c – Chondrogenic cell

29. c – Lung

30. a – Embedded in superficial endometrium

31. d – Multilocular cells

32. c – Give rise to sperms in the male

33. a – Nucleus pulposus

34. a – Central nervous system

35. b – Formation of sperms

36. a – Fertilization

37. d – Double unit membrane

38. d – Third week

39. d – Mast cells

40. d – Hyaline cartilage

41. c – Ampulla of fallopian tube

42. c – Occurs at the embryonic pole

43. a – Decidua

44. b – Medulla of suprarenal gland

45. a – Formation of trilaminar germ disc

46. b – Manubriosternal joint

47. a – Microfilaments

48. b – Formation of bilaminar germ disc

49. c – Epiphyseal

50. c – Multilobed

51. c – Apical ligament

52. a – Spermatogenesis begins at birth

53. a – Removal of glycoprotein coat from sperm head

54. b – Kidneys and gonads

55. a – Forms adhesions between different parts of the embryo

56. b – Muscles of the limbs

57. c – Fibrous

58. a – Daughter cells of pluripotent stem cells

59. d – Dark brown network by silver staining

60. b – Osteoclast

61. c – Second part of the duodenum

62. a – Stomach

63. b – Fascia

64. a – 6th day of gestation

****



Formative 15 “PMS” with answers

Prepared by group “A”

---

1. Vessels that carry blood from the heart to the periphery are which of the following?
    a. Capillaries
    b. Arteries
    c. Sinusoids
    d. Veins

****

2. What are the 3 types of muscles?
    a. Tendons, joints, ligaments
    b. Skeletal, smooth, stringy
    c. Cardiac, skeletal, smooth
    d. Flexion, extension, smooth

****

3. What is the type of collagen fiber in bone?
    a. Type I
    b. Type II
    c. Type III
    d. Type IV

****

4. Fertilization occurs at which of the following?
    a. Infundibulum of fallopian tube
    b. Isthmus of fallopian tube
    c. Intramural part of the Fallopian tube
    d. Ampulla of fallopian tube

****

5. Which of the following is a part of the digestive system?
   a. Lung
   b. Trachea
   c. Nose
   d. Stomach

****

6. Vitelline duct connects definitive yolk sac with which of the following?
   a. Midgut
   b. Amnion
   c. Hindgut
   d. Foregut

****

7. Which statement of the following best describes the chorion frondosum?
   a. The part of chorion does not participate in exchange process
   b. Lies opposite decidua parietalis as pregnancy advances
   c. Fetal part of placenta
   d. Increase in size and complexity on the abembryonic pole

****

8. Which of the following is a primordium of the umbilical cord?
   a. Cytotrophoblast
   b. Allantois
   c. Extraembryonic coelom
   d. Connecting stalk

****

9. What is the structure of the microvilli?
   a. Intermediate filaments
   b. Microfilaments
   c. Thick filaments
   d. Microtubules

****

10. Which one of the following is the endometrium of the pregnant uterus?
   a. Amnion
   b. Umbilical cord
   c. Decidua
   d. Chorion

****

11. Which of the following occurs in the 3rd week of pregnancy?
    a. Beginning of gastrulation
    b. Formation of amniotic cavity
    c. Formation of bilaminar germ disc
    d. Formation of yolk sac

****

12. What is the distinguishing character of the reticular fibers?
    a. Dark brown in fresh preparation
    b. Yellowish in fresh preparation
    c. Dark brown network by silver staining
    d. Readily visible by Hx&E preparation

****

13. Which one of the following structures is devoid of the lymph vessels?
    a. Central nervous system
    b. Dermis of the skin
    c. External ear
    d. Face

****

14. A 45-year-old woman states that she has experienced pain for 2 years over her left lower back; pain that radiates to her left lower limb. Radiologic examination revealed intervertebral disc herniation. What is the type joint the intervertebral disc belong to?
    a. Synovial joint
    b. Fibrous joint
    c. Secondary cartilaginous joint
    d. Primary cartilaginous joint

****

15. Which statement of the following describes capacitation of the sperm?
    a. Prevents polyspermy
    b. It is caused by the zona pellucida
    c. Means removal of glycoprotein coat from the sperm head
    d. Means removal of the tail

****

16. What is the stabilizing cell structure that resists tension?
    a. Intermediate filaments
    b. Myosin filaments
    c. Microfilaments
    d. Microtubules

****

17. Which of the following is an anomaly in the position of the placenta?
    a. Placenta previa
    b. Succenturiate placenta
    c. Placenta tripartite
    d. Membranous placenta

****

18. A 32-year-woman is admitted to the emergency department as a rape case. Fluids from her vagina are collected for sperm examination. Which of the following male organs is responsible for sperm production?
    a. Seminal vesicles
    b. Kidneys
    c. Testis
    d. Prostate gland

****

19. What is the dividing cell in appositional growth?
    a. Chondroblast
    b. Chondrogenic
    c. Chondrocyte
    d. Osteocyte

****

20. Which of the following is an irregular type of bone?
    a. Radius
    b. Patella
    c. Vertebrae
    d. Scapula

****

21. Which of the following are developed from lateral plate mesoderm?
    a. Mucous membrane of the gastro-intestinal tract
    b. Mucous membrane of part of urinary bladder
    c. Muscles of the limbs
    d. Mucous membrane of the respiratory tract

****

22. Which of the following best describes anatomy of the kidney?
    a. Adult kidneys lie in the pelvis
    b. The suprarenal gland covers its lower pole
    c. The left kidney is lower than the right kidney
    d. The ureter conducts the urine from the kidney to the urinary bladder

****

23. Which of the following means gastrulation?
    a. Formation of bilaminar germ disc
    b. Formation of primitive streak
    c. Formation of trilaminar germ disc
    d. Embryonic folding

****

24. What is the distinguishing character of the hemidesmosomes?
    a. Result in sealing between adjacent cells
    b. Present between adjacent epithelial cells
    c. Help rapid conduction of nerve impulses
    d. Present between cells and the underlying basement membrane

****

25. Which is the bone-forming cell?
    a. Osteoclast
    b. Osteocyte
    c. Osteogenic
    d. Osteoblast

****

26. What is the normal amount of the amniotic fluid?
    a. Three liters
    b. Six liters
    c. One liter
    d. Half a liter

****

27. The normal position of the uterus is which of the following?
    a. Retroverted retroflexed
    b. Anteverted anteflexed
    c. Retroverted
    d. Retroflexed

****

28. What is the normal organization of the urinary system?
    a. Urethra, ureters, kidneys, urinary bladder
    b. Ureters, urinary bladder, kidneys, urethra
    c. Kidneys, ureters, urinary bladder, urethra
    d. Urinary bladder, kidneys, urethra, ureters

****

29. What is the male external genital organ of the following?
    a. Epididymis
    b. Ejaculatory ducts
    c. Ductus deferens
    d. Scrotum

****

30. Which of the following is a function of the amnion?
    a. Show
    b. Oligohydramnios
    c. Adhesions
    d. Hydramnios

****

31. Intermediate mesoderm will give rise to which of the following?
    a. Neural tube
    b. Somites
    c. Gonads
    d. Heart

****

32. Which statement of the following does not describe spermatogenesis?
    a. Takes place in the testis
    b. Spermatogenesis begins at birth
    c. It ends at adult age
    d. Primary spermatocytes undergo mitotic division to form spermatids

****

33. Neural tube will give rise to which of the following?
    a. Heart
    b. Central nervous system
    c. Kidneys and gonads
    d. Muscles

****

34. Intermediate mesoderm will give rise to which of the following?
    a. Neural tube
    b. Heart
    c. Somites
    d. Kidneys and gonads

****

35. What is the distinguishing character of progenitor cells?
    a. Are stem cells
    b. Divide infrequently
    c. Slowly dividing
    d. Are daughter cells of pleuripotent stem cells

****

36. What is the distinguishing character of epithelial tissues?
    a. Abundant extracellular matrix
    b. Numerous blood vessels
    c. Closely aggregated cells
    d. Widely separated cells

****

37. In which cell is the ribosome absent?
    a. Lymphocytes
    b. Kidney cells
    c. Erythrocytes
    d. Muscle cells

****

38. Which of the following occurs in the 2nd week of pregnancy?
    a. Embryonic folding
    b. Beginning of gastrulation
    c. Formation of primitive streak
    d. Formation of bilaminar germ disc

****

39. A 32-year-woman is admitted to the emergency department as a rape case. Fluids from her vagina are collected for fructose examination. Which of the following male organs is responsible for fructose production?
    a. Seminal vesicles
    b. Testis
    c. Prostate gland
    d. Kidneys

****

40. The blastocyst begins implantation at which day?
    a. 10th day of gestation
    b. 3rd day of gestation
    c. 6th day of gestation
    d. 8th day of gestation

****

41. Which statement best describes the superficial fascia?
    a. Membranous layer of connective tissue
    b. Mixture of loose areolar and adipose tissue
    c. May determine the path taken by pathogenic organisms
    d. Retinacula; one of its varieties

****

42. Which cell synthesizes the structural unit of collagen (tropocollagen)?
    a. Fibrocyte
    b. Fibroblast
    c. Plasma cell
    d. Fat cell

****

43. What is the structure of the nuclear envelope?
    a. Unit membrane and ribosomes
    b. Unit membrane with cristae
    c. Double unit membrane
    d. Acidophilic membrane

****

44. The heart is surrounded by:
    a. Meninges
    b. Pleura
    c. Peritoneum
    d. Pericardium

****

45. Meiosis II of secondary oocyte is completed at:
    a. Fertilization
    b. Ovulation
    c. Intrauterine life
    d. Before ovulation

****

46. The central nervous system is surrounded by:
    a. Pericardium
    b. Pleura
    c. Meninges
    d. Peritoneum

****

47. Which of the following is part of the respiratory system?
    a. Kidney
    b. Lung
    c. Stomach
    d. Esophagus

****

48. Which of the following is responsible for formation of the trophoblast?
    a. Zona pellucida
    b. Outer cell mass
    c. Blastocyst cavity
    d. Inner cell mass

****

49. A 5-year-old boy had been playing with a small toy car. Soon after, he put a wheel from one of the cars in his mouth and began coughing. Where in the tracheobronchial tree is the common site for a foreign object to lodge?
    a. The left tertiary bronchus
    b. The beginning of trachea
    c. The left primary bronchus
    d. The right primary bronchus

****

50. A 3-year-old boy was brought to the pediatrician with delayed growth and short stature. What type of bone growth defect is present in this boy?
    a. Appositional
    b. Epiphyseal
    c. Interstitial and appositional
    d. Interstitial

****

51. Which statement describes implantation?

a. Begins about day 11 after fertilization
b. The blastocyst is embedded in the myometrium
c. Occurs at the embryonic pole
d. Usually completed at the end of 3rd week of pregnancy

****

52. A male born baby was brought to the pediatrician due to persistent low temperature. After examination and investigations he was diagnosed as deficiency in thermogenin produced by brown fat. What is the distinguishing character of the brown fat cells?

a. Numerous in adult
b. Multilocular cells
c. Unilocular cells
d. Fat is pigmented

****

53. Union of duct of vas deferens and duct of seminal vesicle gives which of the following?

a. Epididymis
b. Ejaculatory duct
c. Prostate
d. Penile urethra

****

54. What is the lining of absorptive membranes?

a. Columnar cells with microvilli
b. Ciliated cells
c. Cells containing numerous secretory granules
d. Simple squamous epithelium

****

55. What is the part of the sperm responsible for energy production:

a. Midpiece
b. Acrosome
c. Tail
d. Head

****

56. Which statement best describes dizygotic twins?

a. Splitting at two-cell stage,two embryos will have separated placenta.
b. Occurs when one fertilized ovum splits into two or more embryos.
c. May be of the same or different sex
d. Splitting at two cell stage,two embryos will have separated chorionic sac .

****

57. Which of the following are developed from notochord?
a. Mucous membranes of the respiratory tract
b. Mucous membranes of part of urinary bladder
c. Muscles of the limbs
d. Apical ligament

****

58. What is the type of wrist joint?
a. Ball and socket synovial joint
b. Ellipsoid synovial joint
c. Plane synovial joint
d. Pivot synovial joint

****

59. Primary sexual organ of female genital system:
a. Uterus
b. Vulva
c. Vagina
d. Ovary

****

60. Which statement does not describe cleavage?
a. Occurs while the zygote is passing through the fallopian tube to the uterus.
b. Results in an increase in the size of the cell in the zygote.
c. Series of meiotic divisions
d. Its cells are called somitomeres

****

61. Length of each uterine tube:
a. 15 cm
b. 20 cm
c. 10 cm d. 5 cm

****

62. Which does not describe a stage of fertilization?
    a. Release of the mature ovarian follicle from surface of the ovary.
    b. Formation of sperms
    c. Shedding out of cytoplasm
    d. Capacitation

****

63. Which statement describes implantation?
    a. Completed at the end of 3rd week of pregnancy.
    b. Begins about day 11 after fertilization.
    c. Occurs at the abembryonic pole.
    d. The blastocyst is embedded in superficial layers of the endometrium.

****

64. Primitive streak first appears at beginning of:
    a. First week
    b. Fourth week
    c. Third week
    d. Second week

****

65. Type of collagen fiber in compact bone:
    a. Type II
    b. Type VII
    c. Type I
    d. Type III

****

66. Which is not a content of full-term umbilical cord?
    a. Wharton’s jelly
    b. Umbilical arteries
    c. Left umbilical vein
    d. Right umbilical vein

****

67. Which of the following are developed from mesoderm?
    a. Mucous membranes of part of urinary bladder
    b. Muscles of the limbs
    c. Mucous membranes of the gastrointestinal tract
    d. Mucous membrane of the respiratory tract

****

68. Which of the following occurs in the 2nd week of pregnancy?
    a. Formation of the neural tube
    b. Embryonic folding
    c. Formation of the three germ layers
    d. Formation of bilaminar germ disc

****

69. Maternal part of placenta is:
    a. Chorion frondosum
    b. Decidua basalis
    c. Decidua parietalis
    d. Decidua capsularis

****

70. Gomphosis is an example of:
    a. Secondary cartilaginous joint
    b. Fibrous joint
    c. Primary cartilaginous joint
    d. Synovial joint

****

71. What does transitional epithelium lines:
    a. Ureter
    b. Cornea of the eye
    c. Esophagus
    d. Trachea

****

72. What is the site of storage and maturation of sperms:
    a. Seminal vesicles
    b. Efferent ductules
    c. Prostatic gland
    d. Epididymis

****

73. Which statement relates to skeletal muscle?
    a. Attachment which moves the most is the origin
    b. Degree of mobility between origin and insertion may not be reversed .
    c. Its nerve supply is mainly sensory
    d. Attached to the skeleton

****

74. Metachromatic granules are present in:
    a. Mast cells
    b. Fat cells
    c. Pigment cells
    d. Plasma cells

****

75. The heart distributes blood to the body through:
    a. Coronary arteries
    b. Pulmonary veins
    c. Aorta
    d. Pulmonary trunk

****

76. Which of the following are developed from endoderm?
    a. Kidneys
    b. Mucous membranes of the gastrointestinal tract
    c. Epidermis of the skin
    d. Dermis of the skin

****

77. Primary oocyte remains in first meiotic division until:
    a. Before ovulation
    b. Third month of intra uterine life
    c. Menopause
    d. Birth

****

78. Which is a congenital anomaly of the amnion?
    a. Forms sac of amnion
    b. Protection against external trauma
    c. Allows fetal movements
    d. Forms adhesions between different parts of the embryo

****

79. The main pancreatic duct opens into:
    a. Second part of the duodenum
    b. Third part of the duodenum
    c. Fourth part of the duodenum
    d. First part of the duodenum

****

80. Which is developed from notochord?

a. Nucleus polyposis of intervertebral disc
b. Annulus fibrosus of intervertebral disc
c. Brain
d. Spinal cord

****

81. Which of the following cells is multinucleated?
a. Osteocyte
b. Macrophage
c. Osteoblast
d. Osteoclast

****

82. A 35-year-old obese man is admitted to the hospital with jaundice and complaints of abdominal pain. Physical examination reveals an epigastric pain. Radiographic examination reveals multiple gallstones. Which of the following structures is most likely obstructed by the gallstones?
a. Left hepatic duct
b. Common bile duct
c. Right hepatic duct
d. Cystic duct

****

83. Which statement best describes primordial germ cells?
a. Give rise to ova in the female
b. Originated in the wall of the amniotic cavity
c. Give rise to testes in the male
d. Ectodermal in origin

****

84. What is the distinguishing character of the hyaline cartilage matrix?
a. Calcified matrix
b. Contains collagen type II
c. Contains collagen type I
d. Vascular

****

85. Osteon (Haversian system) is present in:
a. Cancellous bone
b. Woven bone
c. Compact bone
d. Spongy bone

****

86. The second envelope of the body is:
   a. Deep fascia
   b. Muscles
   c. Fascia
   d. Skin

****

87. Ribs and sternum are parts of:
   a. Axial skeleton
   b. Bones of the lower limb
   c. Bones of the upper limb
   d. Appendicular skeleton

****

88. What is the shape of the nucleus of neutrophils?
   a. Kidney-shaped or indented
   b. Irregular
   c. Rounded
   d. Multilobed

****

89. Which arteries carry deoxygenated blood?
   a. Aorta
   b. Brachial
   c. Coronary
   d. Pulmonary

****

90. Which is derived from neural crest?
   a. Cortex of the suprarenal gland
   b. Cartilage and bone
   c. Medulla of the suprarenal gland
   d. The cardiovascular system

****

91. Which of the following is considered as a secondary cartilaginous joint?
   a. Manubriosternal joint
   b. Shoulder
   c. Suture
   d. Junction between epiphysis and diaphysis

****

92. Which is a constituent of placenta?
    a. Wharton's jelly
    b. Cotyledons
    c. Allantois
    d. Umbilical arteries

****

93. Which cell increases in parasitic diseases?
    a. Basophil
    b. Neutrophil
    c. Macrophage
    d. Eosinophil

****

94. Which statement best describes primordial germ cells?
    a. Give rise to sperms in the male
    b. Give rise to ovary in the female
    c. Ectodermal in origin
    d. Originated in the wall of the amniotic cavity

****

95. Red blood cells are produced in:
    a. Heart
    b. Bone marrow
    c. Lymph nodes
    d. Liver

****

96. The cartilage forms the skeleton of the fetus is:
    a. Elastic cartilage
    b. Fibrocartilage
    c. Hyaline cartilage
    d. Yellow cartilage

****

97. Which is a congenital anomaly of the vitelline duct?
    a. Urachal cyst
    b. Meckel's diverticulum
    c. Urachal fistula
    d. Urachal sinus

****

Answers

1. b. Arteries.

2. c. Cardiac, skeletal, smooth.

3. a. Type I.

4. d. Ampulla of fallopian tube.

5. d. Stomach.

6. a. Midgut.

7. c. Fetal part of placenta.

8. d. Connecting stalk.

9. b. Microfilaments.

10. c. Decidua.

11. a. Beginning of gastrulation.

12. c. Dark brown network by silver staining.

13. a. Central nervous system.

14. c. Secondary cartilaginous joint.

15. c. Means removal of glycoprotein coat from the sperm head.

16. a. Intermediate filaments.

17. a. Placenta previa.

18. c. Testis.

19. b. Chondrogenic.

20. c. Vertebrae.

21. c. Muscles of the limbs.

22. d. The ureter conducts the urine from the kidney to the urinary bladder.

23. c. Formation of trilaminar germ disc.

24. d. Present between cells and the underlying basement membrane.

25. d. Osteoblast.

26. c. One liter.

27. b. Anteverted anteflexed.

28. c. Kidneys → ureters → urinary bladder → urethra.

29. d. Scrotum.

30. a. Show (protection).

31. c. Gonads.

32. b. Spermatogenesis begins at birth.

33. b. Central nervous system.

34. d. Kidneys and gonads.

35. d. Are daughter cells of pleuripotent stem cells.

36. c. Closely aggregated cells.

37. c. Erythrocytes.

38. d. Formation of bilaminar germ disc.

39. a. Seminal vesicles.

40. c. 6th day of gestation.

41. b. Mixture of loose areolar and adipose tissue.

42. b. Fibroblasts.

43. c. Double unit membrane.

44. d. Pericardium.

45. a. Fertilization.

46. c. Meninges.

47. b. Lung.

48. b. Outer cell mass.

49. d. The right primary bronchus.

50. b. Epiphyseal.

51. c. Occurs at the embryonic pole.

52. b. Multilocular cells.

53. b. Ejaculatory duct.

54. a. Columnar cells with microvilli.

55. a. Midpiece.

56. c. May be of the same or different sex.

57. d. Apical ligament.

58. b. Ellipsoid synovial joint.

59. d. Ovary.

60. c.Occurs while the zygote is passing through the fallopian tube to the uterus.

61. c. 10 cm.

62. b.Release of the mature ovarian follicle from surface of the ovary.

63. d. The blastocyst is embedded in superficial layers of the endometrium.

64. c. Third week.

65. c. Type I.

66. d. Right umbilical vein.

67. b. Muscles of the limbs.

68. d. Formation of bilaminar germ disc.

69. b. Decidua basalis.

70. b. Fibrous.

71. a. Ureter.

72. d. Epididymis.

73. d. Attached to the skeleton.

74. a. Mast cells.

75. c. Aorta.

76. b. Mucous membranes of the gastro-intestinal tract.

77. a. Before ovulation.

78. d. Forms adhesions between different parts of the embryo.

79. a. Second part of the duodenum.

80. a. Nucleus polyposis of intervertebral disc.

81. d. Osteoclast.

82. b. Common bile duct.

83. a. Give rise to ova in the female.

84. b. Contains collagen type II.

85. c. Compact bone.

86. d. Fascia.

87. a. Axial skeleton.

88. d. Multilobed.

89. d. Pulmonary.

90. c. Medulla of the suprarenal gland.

91. a. Manubriosternal joint.

92. b. Cotyledons.

93. d. Eosinophil.

94. a. Give rise to sperms in male.

95. b. Bone marrow.

96. c. Hyaline cartilage.

97. b. Meckel’s diverticulum.

Prepared by group “A”

Formative 16 “PMS” with answers

Prepared by group “A”

---

1. Neural tube will give rise to which of the following?
a. Kidneys and gonads
b. Central nervous system
c. Heart
d. Muscles
****

2. A pigment present in muscle cells is which of these?
a. Hemoglobin
b. Melanin
c. Myoglobin
d. Lipofuscin
****

3. Union of duct of vas deferens and duct of seminal vesicle gives which of the following?
a. Epididymis
b. Penile urethra
c. Prostate
d. Ejaculatory duct
****

4. In which cell is the ribosome absent?
a. Kidney cells
b. Lymphocytes
c. Erythrocytes
d. Muscle cells
****

5. A 5-year-old boy had been playing with a small toy car. Soon after, he put a wheel from one of the cars in his mouth and began coughing. Where in the tracheobronchial tree is the common site for a foreign object to lodge?
a. The beginning of the trachea
b. The right primary bronchus
c. The left tertiary bronchus
d. The left primary bronchus
****
6. Fertilization occurs at which of the following?
a. Intramural part of the Fallopian tube
b. Isthmus of Fallopian tube
c. Infundibulum of fallopian tube
d. Ampulla of fallopian tube
****
7. What is the structure of the microvilli?
a. Microfilaments
b. Intermediate filaments
c. Thick filaments
d. Microtubules
****
8. The cartilage forms the skeleton of the fetus is which of the following?
a. Yellow cartilage
b. Fibrocartilage
c. Hyaline cartilage
d. Elastic cartilage
****
9. Where is the osteon (Haversian system) present?
a. Cancellous bone
b. Compact bone
c. Woven bone
d. Spongy bone
****
10. Which one of the following is a congenital anomaly of the amnion?
a. Forms sac of amnion
b. Allows fetal movements
c. Forms adhesions between different parts of the embryo

d. Protection against external trauma
****
11. Which statement of the following describes capacitation of the sperm?
a. Prevents polyspermy
b. Means removal of glycoprotein coat from the sperm head
c. It is caused by the zona pellucida
d. Means removal of the tail
****
12. Which of the following are developed from notochord?
a. Muscles of the limbs
b. Mucus membranes of the respiratory tract
c. Mucus membranes of part of urinary bladder
d. Apical ligament
****
13. Which of the following means gastrulation?
a. Formation of primitive streak
b. Embryonic folding
c. Formation of bilaminar germ disc
d. Formation of trilaminar germ disc
****
14. Which one of the following statements describes the implantation?
a. Begins about day 11 after fertilization
b. Usually completed at the end of 3rd week of pregnancy
c. Occurs at the embryonic pole
d. The blastocyst is embedded in the myometrium
****
15. Which one of the following is the endometrium of the pregnant uterus?
a. Chorion
b. Decidua
c. Umbilical cord
d. Amnion
****
16. Which one of the following is a constituent of the placenta?
a. Wharton’s jelly
b. Allantois

c. Umbilical arteries
d. Cotyledons
****
17. Which of the following occurs in the 2nd week of pregnancy?
a. Formation of primitive streak
b. Beginning of gastrulation
c. Embryonic folding
d. Formation of bilaminar germ disc
****
18. Vessels that carry blood from the heart to the periphery are which of the following?
a. Sinusoids
b. Veins
c. Capillaries
d. Arteries
****
19. What is the dividing cell in appositional growth?
a. Chondrocyte
b. Chondrogenic
c. Osteocyte
d. Chondroblast
****
20. The central nervous system is surrounded by which of the following?
a. Peritoneum
b. Meninges
c. Pericardium
d. Pleura
****
21. The heart is surrounded by which of the following?
a. Meninges
b. Pericardium
c. Pleura
d. Peritoneum
****
22. Myoepithelium is an example of epithelial tissue specialized for which of this function?
a. Contractile function

b. Sensory reception
c. Filtration
d. Secretion
****
23. Which of followings is an irregular type of bones?
a. Radius
b. Vertebrae
c. Patella
d. Scapula
****
24. A 3-year-old boy was brought to the pediatrician with delayed growth and short stature. What type of bone growth defect is present in this boy?
a. Epiphyseal
b. Interstitial and appositional
c. Appositional
d. Interstitial
****
25. Gomphosis is an example of which type of the joints?
a. Fibrous
b. Secondary cartilaginous
c. Primary cartilaginous
d. Synovial
****
26. What is the maternal part of the placenta?
a. Decidua basalis
b. Decidua parietalis
c. Chorion frondosum
d. Decidua capsularis
****
27. Which of the following are developed from mesoderm?
a. Mucus membranes of the respiratory tract
b. Mucus membranes of part of urinary bladder
c. Mucous membranes of the gastro-intestinal tract
d. Muscles of the limbs
****

28. Which cell is increased in parasitic diseases?
a. Eosinophil
b. Neutrophil
c. Microphage
d. Basophil
****
29. What is the stabilizing cell structure that resists tension?
a. Microfilaments
b. Intermediate filaments
c. Myosin filaments
d. Microtubule
****
30. Which of the following does not describe a stage of fertilization?
a. Release of the mature ovarian follicle from surface of the ovary
b. Shedding out of cytoplasm
c. Formation of sperms
d. Capacitation
****
31. Which of the following occurs in the 3rd week of pregnancy?
a. Formation of amniotic cavity
b. Formation of bilaminar germ disc
c. Beginning of gastrulation
d. Formation of yolk sac
****
32. What is the lining of the absorptive membranes?
a. Simple squamous epithelium
b. Ciliated cells
c. Columnar cells with microvilli
d. Cells containing numerous secretory granules
****
33. Type of collagen fiber in compact bone is which of the following?
a. Type I
b. Type III
c. Type II
d. Type VI
****

34. What is the structure of the nuclear envelope?
a. Double unit membrane
b. Acidophilic membrane
c. Unit membrane with cristae
d. Unit membrane and ribosomes
****
35. Primary oocyte remains in 1st meiotic division stage till which of the following?
a. 3rd month of intra uterine life
b. Birth
c. Before ovulation
d. Menopause
****
36. What is the statement is related to the skeletal muscles?
a. Its nerve supply is mainly sensory
b. Attached to the skeleton
c. Attachment which moves the most is the origin
d. Degree of mobility between origin and insertion may not be reversed
****
37. A 35-year-old obese man is admitted to the hospital with jaundice and complaints of abdominal pain. Physical examination reveals an epigastric pain. Radiographic examination reveals multiple gallstones. Which of the following structures is most likely obstructed by the gallstones?
a. Common bile duct
b. Right hepatic duct
c. Left hepatic duct
d. Cystic duct
****
38. What is the male external genital organ of the following?
a. Ejaculatory ducts
b. Epididymis
c. Ductus deferens
d. Scrotum
****
39. What is the structure of the nuclear envelope?
a. Acidophilic membrane

b. Double unit membrane
c. Unit membrane and ribosomes
d. Unit membrane with cristae
****
40. Primitive streak first appears at beginning of which of the following weeks?
a. Second week
b. First week
c. Third week
d. Fourth week
****
41. The pars fibrosa of the nucleolus contains which of the following?
a. Only carbohydrates
b. Ribosomal subunits
c. Dense lipid droplets
d. Chromatin filaments and filamentous RNA
****
42. What is the structure of the microvilli?
a. Microfilaments
b. Intermediate filaments
c. Microtubules
d. Thick filaments
****
43. Which of the following are developed from endoderm?
a. Mucous membranes of the gastro-intestinal tract
b. Kidneys
c. Dermis of the skin
d. Epidermis of the skin
****
44. Intermediate mesoderm will give rise to which of the following?
a. Neural tube
b. Heart
c. Somites
d. Gonads
****
45. What is the length of each of the uterine tubes?
a. 15 cm

b. 5 cm
c. 10 cm
d. 20 cm
****
46. Which statement of the following best describes the primordial germ cells?
a. Give rise to sperms in the male
b. Ectodermal in origin
c. Give rise to ovary in the female
d. Originated in the wall of the amniotic cavity
****
47. Type of collagen fiber in compact bone is
a. Type VII
b. Type II
c. Type I
d. Type III
****
48. A 32-year-woman is admitted to the emergency department as a rape case. Fluids from her vagina are collected for fructose examination. Which of the following male organs is responsible for fructose production?
a. Prostate gland
b. Testis
c. Seminal vesicles
d. Kidneys
****
49. A 32-year-woman is admitted to the emergency department as a rape case. Fluids from her vagina are collected for sperm examination. Which of the following male organs is responsible for sperm production?
a. Testis
b. Kidneys
c. Seminal vesicles
d. Prostate gland
****
50. Which of the following is an anomaly in the position of the placenta?
a. Membranous placenta
b. Placenta tripartite

c. Placenta previa
d. Succenturiate placenta
****
51. The basal lamina is described as PAS +ve, which indicates the presence of which of the following?
a. Nucleic acids
b. Glycoproteins
c. Lipids
d. Keratin
****
52. What are the 3 types of muscles?
a. Cardiac, skeletal, smooth
b. Tendons, joints, ligaments
c. Flexion, extension, smooth
d. Skeletal, smooth, stringy
****
53. What is the type of wrist joint?
a. Ball and socket synovial joint
b. Plane synovial joint
c. Pivot synovial joint
d. Ellipsoid synovial joint
****
54. Which of the following arteries carry deoxygenated blood?
a. Pulmonary
b. Coronary
c. Brachial
d. Aorta
****
55. Which one of the following structures is devoid of the lymph vessels?
a. Face
b. External ear
c. Dermis of the skin
d. Central nervous system
****
56. Meiosis II of secondary oocytes is completed at which of the following?
a. Intrauterine life

b. Ovulation
c. Before ovulation
d. Fertilization
****
57. Which of the followings is a congenital anomaly of the vitelline duct?
a. Uracheal sinus
b. Mickel's diverticulum
c. Uracheal cyst
d. Uracheal fistula
****
58. Which is the statement best describes dizygotic twins of the following?
a. Occurs when one fertilized ovum splits into two or more embryos
b. Splitting at two cell stage; two embryos will have separated chorionic sac
c. Splitting at two cell stage; two embryos will have separated placenta
d. May be of the same or different sex
****
59. Which statement of the following best describes the chorion frondosum?
a. The part of chorion does not participate in exchange process
b. Lies opposite decidua parietalis as pregnancy advance
c. Increase in size and complexity on the abembryonic pole
d. Fetal part of placenta
****
60. What is the type of collagen fiber in bone?
a. Type III
b. Type II
c. Type I
d. Type IV
****
61. What is the dividing cell in appositional growth?
a. Chondroblast
b. Osteocyte
c. Chondrogenic
d. Chondrocyte
****

62. What cytoskeletal filament is abundant in microvilli?
a. Actin filaments
b. Intermediate filaments
c. Microtubules
d. Myosin filaments
****
63. Which of the following is a function of the amnion?
a. Oligohydramnios
b. Hydramnios
c. Show
d. Adhesions
****
64. Which one of the following statements describes the implantation?
a. Occurs at the abembryonic pole
b. Begins about day 11 after fertilization
c. Completed at the end of 3rd week of pregnancy
d. The blastocyst is embedded in superficial layers of the endometrium
****
65. Microvilli are mainly found in epithelia specialized for which of these?
a. Absorption
b. Contraction
c. Secretion
d. Protection
****
66. What is the second envelope of the body?
a. Skin
b. Muscles
c. Deep fascia
d. Fascia
****
67. Where are the metachromatic granules present?
a. Fat cells
b. Pigment cells
c. Mast cells
d. Plasma cells
****

68. Which of the following is a part of the digestive system?
a. Nose
b. Trachea
c. Stomach
d. Lung
****
69. Which is the bone forming cell?
a. Osteocyte
b. Osteoclast
c. Osteoblast
d. Osteogenic
****
70. Which of the following is responsible for formation of the trophoblast?
a. Zona pellucida
b. Blastocyst cavity
c. Outer cell mass
d. Inner cell mass
****
71. What is the part of the sperm responsible for energy production of the following?
a. Midpiece
b. Head
c. Tail
d. Acrosome
****
72. Vitelline duct connects definitive yolk sac with which of the following?
a. Hindgut
b. Midgut
c. Foregut
d. Amnion
****
73. The nuclear envelope is formed of which of the following?
a. Two parallel unit membranes
b. Continuous membrane without pores
c. A non-selective barrier
d. A single lipid layer
****

74. In which cell is the ribosome absent?
a. Muscle cells
b. Erythrocytes
c. Kidney cells
d. Lymphocytes
****
75. What does transitional epithelium line?
a. Ureter
b. Esophagus
c. Trachea
d. Cornea of the eye
****
76. Which statement of the following best describes the superficial fascia?
a. May determine the path taken by pathogenic organisms
b. Mixture of loose areolar and adipose tissues
c. Retinacula; one of its varieties
d. Membranous layer of connective tissue
****
77. Which of the following is a part of the respiratory system?
a. Esophagus
b. Lung
c. Kidney
d. Stomach
****
78. What is the site of storage and maturation of the sperms of the following?
a. Seminal vesicles
b. Prostatic gland
c. Epididymis
d. Efferent ductules
****
79. Which cell synthesize the structural unit of collagen (tropocollagen)?
a. Fat cell
b. Fibroblasts
c. Plasma cells
d. Fibrocyte

****
80. Which of the following are developed from lateral plate of mesoderm?
a. Muscles of the limbs
b. Mucus membrane of the respiratory tract
c. Mucous membrane of the gastro-intestinal tract
d. Mucus membrane of part of urinary bladder
****
81. A 45-year-old woman states that she has experienced pain for 2 years over her left lower back; pain that radiates to her left lower limb. Radiologic examination revealed intervertebral disc herniation. What is the type joint the intervertebral disc belong to?
a. Fibrous joint
b. Synovial joint
c. Secondary cartilaginous joint
d. Primary cartilaginous joint
****
82. What is the normal organization of the urinary system?
a. Urethra, ureters, kidneys, urinary bladder
b. Kidneys, ureters, urinary bladder, urethra
c. Urinary bladder, kidneys, urethra, ureters
d. Ureters, urinary bladder, kidneys, urethra
****
83. Which statement of the following does not describe spermatogenesis?
a. Takes place in the testis
b. It ends at adult age
c. Primary spermatocytes undergo mitotic division to form spermatids
d. Spermatogenesis begins at birth
****
84. A male born baby was brought to the pediatrician due to persistent low temperature. After examination and investigations he was diagnosed as deficiency in thermogenin produced by brown fat. What is the distinguishing character of the brown fat cells?
a. Unilocular cells
b. Numerous in adult
c. Fat is pigmented

d. Multilocular cells
****
85. What is the shape of the nucleus of neutrophils?
a. Kidney shaped or indented
b. Multilobed
c. Irregular
d. Rounded
****
86. A male born baby was brought to the pediatrician due to persistent low temperature. After examination and investigations he was diagnosed as deficiency in thermogenin produced by brown fat. What is the distinguishing character of the brown fat cells?
a. Fat is pigmented
b. Unilocular cells
c. Multilocular cells
d. Numerous in adult
****
87. Which of the following is a primordium of the umbilical cord?
a. Extraembryonic coelom
b. Connecting stalk
c. Allantois
d. Cytotrophoblast
****
88. Which of the following statements does not describe the cleavage?
a. Its cells are called somitomers
b. Series of meiotic divisions
c. Occurs while the zygote is passing through the fallopian tube to the uterus
d. Results in an increase in the size of the cells in the zygote
****
89. Ribs and sternum are parts of which of the following?
a. Bones of the lower limb
b. Bone of the upper limb
c. Axial skeleton
d. Appendicular skeleton
****

90. The heart distributes blood to the whole body through which of the following?
a. Pulmonary veins
b. Pulmonary trunk
c. Coronary arteries
d. Aorta
****
91. What is the primary sexual organ of the female genital system of the following?
a. Vulva
b. Ovary
c. Uterus
d. Vagina
****
92. Intermediate mesoderm will give rise to which of the following?
a. Neural Tube
b. Heart
c. Somites
d. Kidneys and gonads
****
93. What is the distinguishing character of the Reticular fibers?
a. Yellowish in fresh preparation
b. Dark brown network by silver staining
c. Readily visible by Hx & E preparation
d. Dark brown in fresh preparation
****
94. Which of the following are developed from notochord?
a. Brain
b. Annulus fibrosus of intervertebral disc
c. Spinal cord
d. Nucleus polyposis of intervertebral disc
****
95. The blastocyst begins implantation at which of the following?
a. 8th day of gestation
b. 6th day of gestation
c. 10th day of gestation
d. 3rd day of gestation

****
96. The normal position of the uterus is which of the following?
a. Retroflexed
b. Anteverted anteflexed
c. Retroverted
d. Retroverted retroflexed
****
97. What does transitional epithelium line?
a. Cornea of the eye
b. Trachea
c. Ureter
d. Esophagus
****
98. Which is the bone forming cell?
a. Osteocyte
b. Osteoclast
c. Osteoblast
d. Osteogenic
****
99. Where are the metachromatic granules present?
a. Pigment cells
b. Plasma cells
c. Mast cells
d. Fat cells
****
100. Which of the following best describes anatomy of the kidney?
a. The suprarenal gland covers its lower pole
b. Adult kidneys lie in the pelvis
c. The left kidney is lower than the right kidney
d. The ureter conducts the urine from the kidney to the urinary bladder
****
101. Carotenoid pigments are an example of which of these?
a. Crystalline pigments
b. Hematogenous pigments
c. Exogenous pigments
d. Endogenous
****

102. The cartilage forms the skeleton of the fetus is:
a. Fibrocartilage
b. Hyaline cartilage
c. Yellow cartilage
d. Elastic cartilage
****
103. Which of the following is considered as a secondary cartilaginous joint?
a. Junction between epiphysis and diaphysis
b. Suture
c. Manubriosternal joint
d. Shoulder
****
104. Lipofuscin is sometimes called "wear and tear" pigment because of which of the following?
a. Increases in amount with age
b. Is derived from external sources
c. Gives skin its color
d. Is involved in oxygen transport
****
105. Elastic cartilage is characterized by which of the following?
a. No perichondrium
b. Primary skeletal tissue in the fetus
c. Collagen is mainly type II
****
106. What is the type of collagen fiber in bone?
a. Type I
b. Type IV
c. Type II
d. Type III
****
107. What is the normal amount of the amniotic fluid?
a. Three liters
b. One liter
c. Six liters
d. Half a liter
****

108. What is the distinguishing character of the epithelial tissues?
a. Closely aggregated cells
b. Numerous blood vessels
c. Abundant extracellular matrix
d. Widely separated cells
****
109. Which of the following occurs in the 2nd week of pregnancy?
a. Formation of the neural tube
b. Formation of bilaminar germ disc
c. Embryonic folding
d. Formation of the three germ layers
****
110. Which of the following structures is not a content of full-term umbilical cord?
a. Wharton's jelly
b. Umbilical arteries
c. Left umbilical vein
d. Right umbilical vein
****
111. Which of the following is a multinucleated cell?
a. Osteoclast
b. Lymphocyte
c. Hepatocyte
d. Plasma cell
****
112. What is the distinguishing character of the hemidesmosomes?
a. Help rapid conduction of nerve impulses
b. Present between cells and the underlying basement membrane
c. Result in sealing between adjacent cells
d. Present between adjacent epithelial cells
****
113. Where are produced red blood cells?
a. Bone marrow
b. Liver
c. Lymph nodes
d. Heart
****

114. The main pancreas duct opens into which of the following?
a. Second part of the duodenum
b. First part of the duodenum
c. Third part of the duodenum
d. Fourth part of the duodenum
****
115. What is the distinguishing character of the progenitor cells?
a. Slowly dividing
b. Are daughter cells of pleuripotent stem cells
c. Divide infrequently
d. Are stem cells
****
116. A 3-year-old boy was brought to the pediatrician with delayed growth and short stature. What type of bone growth defect is present in this boy?
a. Interstitial
b. Epiphyseal
c. Appositional
d. Interstitial and appositional
****
117. Which statement of the following best describes the primordial germ cells?
a. Ectodermal in origin
b. Give rise to ova in the female
c. Originated in the wall of the amniotic cavity
d. Give rise to testes in the male
****
118. Which of the following cells is a multinucleated cell?
a. Osteoclast
b. Macrophage
c. Osteoblast
d. Osteocyte
****
119. Microtubule can be seen in which of the following?
a. Golgi
b. Mitotic spindle
c. Microvilli

d. Nucleus
****
120. Which of the following is derived from the neural crest?
a. The cardiovascular system
b. Medulla of the suprarenal gland
c. Cartilage and bone
d. Cortex of the suprarenal gland
****
121. What is the distinguishing character of the hyaline cartilage matrix?
a. Vascular
b. Calcified matrix
c. Contains collagen type II
d. Contains collagen type I

Answers to formative 16 “PMS”

---

1. b. Central nervous system.

2. c. Myoglobin.

3. d. Ejaculatory duct.

4. c. Erythrocytes.

5. b. The right primary bronchus.

6. d. Ampulla of fallopian tube.

7. a. Microfilaments.

8. c. Hyaline cartilage.

9. b. Compact bone.

10. c. Forms adhesions between different parts of the embryo.

11. b. Means removal of glycoprotein coat from the sperm head.

12. d. Apical ligament.

13. d. Formation of trilaminar germ disc.

14. c. Occurs at the embryonic pole.

15. b. Decidua.

16. d. Cotyledons.

17. d. Formation of bilaminar germ disc.

18. d. Arteries.

19. b. Chondrogenic.

20. b. Meninges.

21. b. Pericardium.

22. a. Contractile function.

23. b. Vertebrae.

24. a. Epiphyseal.

25. a. Fibrous.

26. a. Decidua basalis.

27. d. Muscles of the limbs.

28. a. Eosinophil.

29. b. Intermediate filaments.

30. a. Release of the mature ovarian follicle from surface of the ovary.

31. c. Beginning of gastrulation.

32. c. Columnar cells with microvilli.

33. a. Type I.

34. a. Double unit membrane.

35. c. Before ovulation.

36. b. Attached to the skeleton.

37. a. Common bile duct.

38. d. Scrotum.

39. b. Double unit membrane.

40. c. Third week.

41. d. Chromatin filaments and filamentous RNA.

42. a. Microfilaments.

43. a. Mucous membranes of the gastro-intestinal tract.

44. d. Gonads.

45. c. 10 cm.

46. a. Give rise to sperms in the male.

47. c. Type I.

48. c. Seminal vesicles.

49. a. Testis.

50. c. Placenta previa.

51. b. Glycoproteins.

52. a. Cardiac, skeletal, smooth.

53. d. Ellipsoid synovial joint.

54. a. Pulmonary.

55. d. Central nervous system.

56. d. Fertilization.

57. b. Mickel’s diverticulum.

58. d. May be of the same or different sex.

59. d. Fetal part of placenta.

60. c. Type I.

61. c. Chondrogenic.

62. a. Actin filaments.

63. c. Show.

64. d. The blastocyst is embedded in superficial layers of the endometrium.

65. a. Absorption.

66. d. Fascia.

67. c. Mast cells.

68. c. Stomach.

69. c. Osteoblast.

70. c. Outer cell mass.

71. a. Midpiece.

72. b. Midgut.

73. a. Two parallel unit membranes.

74. b. Erythrocytes.

75. a. Ureter.

76. b. Mixture of loose areolar and adipose tissues.

77. b. Lung.

78. c. Epididymis.

79. b. Fibroblasts.

80. a. Muscles of the limbs.

81. c. Secondary cartilaginous joint.

82. b. Kidneys, ureters, urinary bladder, urethra.

83. d. Spermatogenesis begins at birth.

84. d. Multilocular cells.

85. b. Multilobed.

86. c. Multilocular cells.

87. b. Connecting stalk.

88. c. Occurs while the zygote is passing through the fallopian tube to the uterus.

89. c. Axial skeleton.

90. d. Aorta.

91. b. Ovary.

92. d. Kidneys and gonads.

93. b. Dark brown network by silver staining.

94. d. Nucleus polyposis of intervertebral disc.

95. a. 6th day of gestation.

96. b. Anteverted anteflexed.

97. c. Ureter.

98. c. Osteoblast.

99. c. Mast cells.

100. d. The ureter conducts the urine from the kidney to the urinary bladder.

101. c. Exogenous pigments.

102. b. Hyaline cartilag.

103. c. Manubriosternal joint.

104. a. Increases in amount with age.

105. c. Collagen is mainly type II.

106. a. Type I.

107. b. One liter.

108. a. Closely aggregated cells.

109. b. Formation of bilaminar germ disc.

110. d. Right umbilical vein.

111. a. Osteoclast.

112. b. Present between cells and the underlying basement membrane.

113. a. Bone marrow.

114. a. Second part of the duodenum.

115. b. Are daughter cells of pleuripotent stem cells.

116. b. Epiphyseal.

117. d. Give rise to ova in female.

118. a. Osteoclast.

119. b. Mitotic spindle.

120. b. Medulla of the suprarenal gland.

121. c. Contains collagen type II.
